

Parameters Sorted by ASCII Interface Parameter ID

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
0x00	0x01	0x400	0x2380:1	RF	U16	Current Loop Proportional Gain (Cp).
0x01	0x02	0x401	0x2380:2	RF	U16	Current Loop Integral Gain (Ci).
0x02	0x03	0x402	0x2340	RF	INT16	Current loop programmed value. This current will be used to command the drive when Desired State (0x24) is set to 1. Units: 0.01 A.
0x03	0x04	0x403	0x2203	R*	INT16	Winding A Current. Actual current measured at winding A. Units: 0.01 A.
0x04	0x05	0x404	0x2204	R*	INT16	Winding B Current. Actual current measured at winding B. Units: 0.01 A.
0x05	0x06	0x405	0x2210	R*	INT16	Current Offset A. Offset value applied to the winding A current reading. This offset is calculated by the drive at startup. Units: 0.01 A.
0x06	0x07	0x406	0x2211	R*	INT16	Current Offset B. Offset value applied to the winding B current reading. This offset is calculated by the drive at startup. Units: 0.01 A.
0x07	0x08	0x407	0x2212	R*	INT16	X axis of calculated stator current vector. Units: 0.01 A.
0x08	0x09	0x408	0x2213	R*	INT16	Y axis of calculated stator current vector. Units: 0.01 A.
0x09	0x0A	0x409	0x221A	R*	INT16	Current loop output, X axis of stator space.
0x0A	0x0B	0x40A	0x221B	R*	INT16	Current loop output, Y axis of stator space.
0x0B	0x0C	0x40B	0x2214	R*	INT16	Actual Current, D axis of rotor space. Units: 0.01 A.
0x0C	0x0D	0x40C	0x2215	R*	INT16	Actual Current, Q axis of rotor space. Units: 0.01 A.
0x0D	0x0E	0x40D	0x2216	R*	INT16	Commanded current, D axis of rotor space. Part of the internal current loop calculation. Units: 0.01 A.
0x0E	0x0F	0x40E	0x2217	R*	INT16	Commanded Current, Q axis of rotor space. Part of the internal current loop calculation. Units: 0.01 A.
0x0F	0x10	0x40F	None	R*	INT16	Current Error, D axis of rotor space. Units: 0.01 A.
0x10	0x11	0x410	None	R*	INT16	Current Error, Q axis of rotor space. Units: 0.01 A.
0x11	0x12	0x411	None	R*	INT16	Current Integral Value, D axis of rotor space.
0x12	0x13	0x412	None	R*	INT16	Current Integral Value, Q axis of rotor space.
0x13	0x14	0x413	0x2218	R*	INT16	Current Loop Output, D axis of rotor space.
0x14	0x15	0x414	0x2219	R*	INT16	Current Loop Output, Q axis of rotor space.

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0x15	0x16	0x415	0x221D	R*	INT16	Commanded Motor Current. This is the value that is presently being sent to the current loop. It may come from the programmed value, analog reference, velocity loop, etc. depending on the drive's desired state. Units: 0.01 A.
0x17	0x18	0x417	0x6064	R	INT32	Actual Position. For dual encoder systems this is the load encoder position. See Actual Motor Position (0x32) for motor encoder position. Units: Counts. CANopen objects 0x6064 and 0x6063 hold the same value.
0x18	0x19	0x418	0x6069	R*	INT32	Actual Velocity. Units: 0.01 RPM for Junus, microsteps/s for Stepnet in stepper mode, 0.1 encoder counts/s for all other products.
0x19	0x1A	0x419	0x2310	RF	INT32	Analog reference scaling factor. This value is used to scale the analog reference input voltage to a command that will be used to drive the current or velocity loop (depending on drive state). When in current mode (Desired State (0x24) = 2), the value programmed specifies the amount of current to be commanded when ten volts is applied to the analog input. This value is input in 0.01 A units. For example, to command 12 amps at 10 volts the scaling factor would be 1200. When in velocity mode (Desired State (0x24) = 12), the value programmed specifies the commanded velocity corresponding to ten volts on the input. Units: 0.01 RPM for Junus, microsteps/s for Stepnet in stepper mode, 0.1 encoder counts/s for all other products. When in position mode (Desired State (0x24) = 22 or 32), the value programmed specifies the position (in encoder counts) that the drive will be commanded to for a +10 volt input.
0x1A	0x1B	0x41A	0x2311	RF	INT16	Offset Value applied to analog reference input. Units: mV.
0x1B	0x1C	0x41B	0x2205	R*	INT16	Analog Encoder Sine Input Voltage. Also known as Sine Feedback Voltage. Units: 0.1 mV.
0x1C	0x1D	0x41C	0x2206	R*	INT16	Analog Encoder Cosine Input Voltage. Also known as Cosine Feedback Voltage. Units: 0.1 mV.
0x1D	0x1E	0x41D	0x2200	R*	INT16	Analog Reference Input Voltage. Also known as A/D Reference Input Voltage. Units: mV.
0x1E	0x1F	0x41E	0x2201	R*	INT16	High Voltage A/D Reading. The voltage present on the high-voltage bus. Units: 100 mV.

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0x20	0x21	0x420	0x2202	R*	INT16	Dive temperature A/D Reading. Units: degrees C.
0x21	0x22	0x421	0x2110	RF	INT16	Peak Current Limit. Also known as Boost current on stepper drives. This value cannot exceed the peak current rating of the drive. Units: 0.01 A.
0x22	0x23	0x422	0x2111	RF	INT16	Continuous Current Limit. Also known as Run Current on stepper drives. This value should be less than the User Peak Current Limit. Units: 0.01 A.
0x23	0x24	0x423	0x2112	RF	U16	Time at Peak Current Limit. Also known as Time at Boost Current on stepper drives. Units: mS.
0x24	0x25	0x424	0x2300	RF	U16	Desired State:
						Value Description
						0 Drive disabled.
						1 The current loop is driven by the programmed current value.
						2 The current loop is driven by the analog reference
						3 The current loop is driven by the PWM input.
						4 The current loop is driven by the function generator.
						5 UV current mode.
						6-10 Reserved for future use.
						11 The velocity loop is driven by the programmed velocity value.
						12 The velocity loop is driven by the analog reference.
						13 The velocity loop is driven by the PWM.
						14 The velocity loop is driven by the function generator.
						15-20 Reserved for future use.
						21 The position loop is driven by the trajectory generator.
						22 The position loop is driven by the analog reference.
						23 The position loop is driven by the digital input lines. (Pulse & direction, master encoder, etc.)
						24 The position loop is driven by the function generator.
						25 The position loop is driven by the cam tables.
						26-29 Reserved.
						30 The drive is controlled by the CANopen or EtherCAT interface
						31 The microstepper is driven by the trajectory generator.
						32 Reserved for future use.
						33 The microstepper is driven by the digital input lines.
						34 The microstepper is driven by the function generator.
						35 The microstepper is driven by the cam tables.
						36-39 Reserved for future use.

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						40 The microstepper is controlled by the CANopen or EtherCAT interface.
						41 Reserved
						42 Simple micro-stepping mode. For diagnostic use only.
0x25	0x26	0x425	0x221E	R*	INT16	Limited Motor Current Command. Units: 0.01 A.
0x26	0x27	0x426	0x2313	RF	INT16	Analog Reference Input Deadband. Deadband window value applied to the analog command input. Units: mV.
0x27	0x28	0x427	0x2381:1	RF	U16	Velocity Loop Proportional Gain (Vp).
0x28	0x29	0x428	0x2381:2	RF	U16	Velocity Loop Integral Gain (Vi).
0x29	0x2A	0x429	0x2230	R*	INT32	Velocity Loop Limited Velocity. This is the commanded velocity after it passes through the acceleration and velocity limits and the velocity command filter. The velocity error used by the loop is the difference between the actual velocity and this value.
0x2A	0x2B	0x42A	None	R*	INT32	Velocity Loop Error.
0x2B	0x2C	0x42B	None	R*	INT32	Velocity loop Integral Sum.
0x2C	0x2D	0x42C	0x606B	R*	INT32	Commanded Velocity. Units: 0.01 RPM for Junus, microsteps/s for Stepnet in stepper mode, 0.1 encoder counts/s for all other products.
0x2D	0x2E	0x42D	0x6062	R*	INT32	Commanded Position. Also known as Limited Position in CME. Units: counts.
0x2E	0x2F	0x42E	0x2381:3	RF	U16	Velocity Loop Acceleration Feed Forward. The acceleration command from the trajectory generator is multiplied by this value and the result is added to the velocity loop output.
0x2F	0x30	0x42F	0x2341	RF	INT32	Programmed Velocity Command. Only used in Programmed Velocity Mode (Desired State (0x24) = 11) Units: 0.01 RPM for Junus, microsteps/s for Stepnet in stepper mode, 0.1 encoder counts/s for all other products.
0x30	0x31	0x430	0x2382:1	RF	U16	Position Loop Proportional Gain (Pp).
0x31	0x32	0x431	0x2381:4	RF	INT16	Velocity Loop Shift Value. After the velocity loop is calculated, the result is right shifted this many times to arrive at the commanded current value. This allows the velocity loop gains to have reasonable values for high resolution encoders.

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0x32	0x33	0x432	0x2240	R*	INT32	Actual Motor Position. Also known as Motor Encoder Position. For dual encoder systems, this parameter gives the motor encoder position. For single encoder systems, this is the same as the Actual Position Parameter (0x17) . Units: counts.
0x33	0x34	0x433	0x2382:2	RF	U16	Position Loop Velocity Feed Forward (Vff). The Vff value is multiplied by the Instantaneous Commanded Velocity (0x3B) generated by the trajectory generator. The product is added to the output of the position loop. This gain is scaled by 1/16384. Therefore, setting this gain to 0x4000 (16384) would cause the input velocity to be multiplied by 1.0, and the result added to the output of the position loop.
0x34	0x35	0x434	0x2382:3	RF	U16	Position Loop Acceleration Feed Forward (Aff). The Aff value is multiplied by the Instantaneous Commanded Velocity (0x3B) generated by the trajectory generator. The product is added to the output of the position loop.
0x35	0x36	0x435	0x60F4	R*	INT32	Position Loop Error. The difference between Actual Position (0x17) and Commanded Position (0x2D) . Units: counts.
0x36	0x37	0x436	0x2100	RF	U32	Velocity Loop Acceleration Limit. Used by the velocity loop limiter. Not used when velocity loop is controlled by the position loop. Units: 1000 counts/s ² .
0x37	0x38	0x437	0x2101	RF	U32	Velocity Loop Deceleration Limit. Used by the velocity loop limiter. Not used when velocity loop is controlled by the position loop. Units: 1000 counts/s ² .
0x38	0x39	0x438	0x221C	R*	INT16	Actual Motor Current. This current is calculated based on both the D and Q axis currents. Units: 0.01 A.
0x39	0x3A	0x439	0x2102	RF	U32	Velocity Loop Emergency Stop Deceleration Rate. Units: 1000 counts/s ² .
0x3A	0x3B	0x43A	0x2103	RF	INT32	Velocity Loop Velocity Limit. This value is a limit on the commanded velocity used by the velocity loop. Note that this limit is always in effect. Units 0.1 counts/s.
0x3B	0x3C	0x43B	0x2250	R*	INT32	Instantaneous Commanded Velocity. This velocity is the output of the trajectory generator and is the value by which the position loop's velocity feed forward is multiplied. Units: 0.1 encoder counts/s.
0x3C	0x3D	0x43C	0x2251	R*	U32	Instantaneous Commanded Acceleration. This acceleration is the output of the trajectory generator and is the value by which the position loop's acceleration feed forward is multiplied. Units: 10 encoder counts/s ² .

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0x3D	0x3E	0x43D	0x2122	R*	INT32	Trajectory Destination Position. This is the position that the trajectory generator is using as its destination. Units: encoder counts.
0x3E	0x3F	0x43E	0x2104	RF	INT32	Velocity Window. If the absolute value of the velocity loop error exceeds this, then the velocity window bit in the event status word will be set. Units: 0.1 counts/s.
0x3F	0x40	0x43F	0x2105	RF	U16	Velocity Window Time. The velocity window bit in the event status will be cleared when the absolute velocity error is less than the velocity window for this amount of time. Units: ms.
0x40	0x41	0x440	0x2383:1	F	U16	Motor Type. The type of motor connected to the drive. Bit-mapped as follows:
						Bits Description
						0 Set for linear motor, clear for rotary.
						1-3 Reserved.
						4-5 Motor architecture:
						0 Not specified.
						1 Brushed servo motor.
						2 Microstepper.
						3 Brushless servo motor.
						6-15 Reserved.
0x41	0x42	0x441	0x6404	F	String	Motor Manufacturer.
0x42	0x43	0x442	0x6403	F	String	Motor Model.
0x43	0x44	0x443	0x2383:27	F	INT16	Motor Units. This is only used by CME for display. (0=metric, 1=English).
0x44	0x45	0x444	0x2383:9	F	INT32	Motor Inertia (Mass). Units: Rotary = 0.000001 Kg/cm ² ; Linear = 0.0001 Kg.
0x45	0x46	0x445	0x2383:2	F	INT16	Motor Poll Pairs (used only for rotary motors). Number of motor pole pairs (electrical phases) per rotation. For stepper motors, Poll Pairs = (360 deg / Motor deg/step) / 4.
0x46	0x47	0x446	0x2383:16	F	U16	Motor Brake Type. 0=present, 1=none.
0x47	0x48	0x447	0x2383:15	F	U16	Motor Temperature Sensor Type. 0=none, 1=present.
0x48	0x49	0x448	0x2383:12	F	INT32	Motor Torque Constant. Units: 0.00001 Nm/A.
0x49	0x4A	0x449	0x2383:7	F	INT16	Motor Resistance. Units: 10 mΩ.
0x4A	0x4B	0x44A	0x2383:8	F	INT16	Motor Inductance. Units: 10 μH.
0x4B	0x4C	0x44B	0x2383:13	F	INT32	Motor Peak Torque. Units: 0.00001 Nm units.

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0x4C	0x4D	0x44C	0x2383:14	F	INT32	Motor Continuous Torque. Units: 0.00001 Nm units.
0x4D	0x4E	0x44D	0x2383:11	F	INT32	Motor Max Velocity. Units: 0.1 encoder counts/s.
0x4E	0x4F	0x44E	0x2383:3	F	U16	Motor Wiring. 0=standard, 1= drive's U and V outputs are swapped.
0x4F	0x50	0x44F	0x2383:6	RF	INT16	Motor Hall Offset. Offset angle to be applied to the Hall sensors. Units: degrees.
0x50	0x51	0x450	0x2383:4	F	INT16	Motor Hall Type. The type of Hall effect sensors attached to the motor:
						Value Description
						0 No Hall sensors available.
						1 Digital Hall sensors.
						2 Analog Hall sensors.
0x52	0x53	0x452	0x2383:5	F	INT16	Motor Hall Wiring. Bit-mapped as follows:
						NOTE: When analog Halls are used, only bit 8 is relevant.
						Bits Description
						0-2 The Hall wiring code (see below).
						Value Hall Ordering
						0 U V W
						1 U W V
						2 V U W
						3 V W U
						4 W V U
						5 W U V
						6, 7 Reserved
						3 Reserved.
						4 Invert W Hall input if set. Inversion occurs after Halls wiring is changed by bits 0-2.
						5 Invert V Hall input if set. Inversion occurs after Halls wiring is changed by bits 0-2.
						6 Invert U Hall input if set. Inversion occurs after Halls wiring is changed by bits 0-2.
						7 Reserved.
						8 Reverse analog Halls if set.
						9-15 Reserved.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
0x53	0x54	0x453	0x2383:17	F	U16	Motor Brake Activation Time. Units: ms.
0x54	0x55	0x454	0x2383:18	F	U16	Motor Brake Delay Time. After the brake output is activated, the drive will stay enabled for this amount of time to allow the brake to engage. Units: ms.
0x55	0x56	0x455	0x2383:19	F	INT32	Motor Brake Activation Velocity. Also known as Motor Brake Velocity (CANopen). During the Motor Brake Activation Time (0x53) , if the motor's actual velocity falls below this value the brake output is activated immediately. Units: 0.1 counts/s.
0x56	0x57	0x456	0x2383:10	F	U32	Motor Back EMF Constant. Back EMF velocity estimation can be disabled by setting to zero. Units: rotary motor: 0.01 V/Krpm; linear motor: 0.01 V/mps.
0x57	0x58	0x457	0x2383:29	F	U32	Microsteps/Motor Rev. This parameter is used in true microstepping mode. Units: microsteps.
0x58	0x59	0x458	0x2383:33	F	INT32	Motor Gear Ratio. This parameter may be used to store gear ratio information for dual encoder systems where a gearbox sits between the two encoders. This parameter is not used by the firmware and is supported as a convenience to the CME program. Gear ratio is a ratio of two 16-bit values. The first word gives the number of motor turns and is the numerator. The second word gives the number of position turns and is the denominator.
0x59	0x5A	0x459	0x2107	RF	INT16	Hall Velocity Mode Shift Value. This parameter is only used in Hall velocity mode. It specifies a left shift value for the position and velocity information calculated in that mode.
0x5A	0x5B	0x45A	0x2241	RF	INT16	Encoder output configuration. This parameter determines the source of the buffered encoder output on drives which support it. Bit-mapped as follows:
						Bits Description
						0-1 Mode of operation for encoder output lines.
						0 Output buffered primary encoder (hardware buffering).
						1 Configure pins as secondary encoder input.
						2 Output simulated encoder outputs tracking motor encoder.
						3 Output simulated encoder outputs tracking load encoder.
						8-11 For simulated encoder outputs, these bits configure a scaling value that adjusts the number of encoder output counts for each encoder count on the input.

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							0	No adjustment, 1 count on the encoder is 1 output count.	
							1	Double the number of encoder counts.	
							2	Divide the encoder counts by 2.	
							3	Divide the encoder counts by 4.	
							4	Divide the encoder counts by 8.	
							5	Divide the encoder counts by 16.	
							6	Divide the encoder counts by 32.	
							7	Divide the encoder counts by 64.	
							8	Divide the encoder counts by 128.	
							9	Divide the encoder counts by 256.	
0x5B	0x5C	0x45B	0x2383:32	F	INT32	Load Encoder Resolution. Number of Motor Encoder Units (0x61) per encoder count. Linear motors only. Units: encoder units/count.			
0x5C	0x5D	0x45C	0x2383:31	F	INT16	Load Encoder Direction. 0=normal, 1=reverse.			
0x5D	0x5E	0x45D	0x2383:30	F	U16	Load Encoder Type. This parameter identifies the type of encoder used on the load when running in dual loop mode. The encoding of this parameter has changed over time to support more encoder types than were originally envisioned when the parameter was first defined. Bit 12 of the parameter is used to identify which encoding is active.			
						New encoding supported by Feature Set C (starting with version 2.10), Feature Set D, and Feature Set E:			
						Bits	Meaning		
						0-11	Encoder hardware to use:		
							0-15	Same encoder types as listed below in original encoding.	
							16	Simple analog potentiometer for feedback.	
							17	Reserved for custom encoder.	
							18	Reserved for custom encoder.	
							19	Reserved for custom encoder.	
						12	Always set to identify new encoding.		
13	Linear encoder if set, rotary encoder if clear.								

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						14	If set, then don't use this encoder for position feedback.	
						15	Reserved.	
						Original encoding (bit 12 not set):		
						0-3	Encoder hardware to use:	
							0	No load encoder present.
							1	Primary (differential) quad encoder.
							2	Analog encoder.
							3	Secondary quad encoder from input lines.
							4	Servo tube / analog halls.
							5	Resolver
							11	Endat absolute encoder
							12	SSI serial encoder.
							13	BiSS absolute encoder.
							14	Various absolute encoders made by Sanyo Denki, Panasonic and Harmonic Drives.
							15	Harmonic Drives custom encoder.
						4	Linear encoder if set, rotary encoder if clear.	
						5	If set, then don't use this encoder for position feedback.	
						6-15	Must be zero.	
0x5E	0x5F	0x45E	0x2231	R*	INT32	Load Encoder Velocity. Units: Encoder counts/s		
0x5F	0x60	0x45F	0x2106	RF	9 or 14	Velocity Loop Output Filter. A bi-quad filter which acts on the output of the velocity loop. 9 word parameter, see <i>Velocity Loop Filters</i> in the <i>CME 2 User Guide</i> . 14 word parameter (Plus product only), see Filter Coefficients (p. 73) .		
0x60	0x61	0x460	0x2383:20	F	U16	Motor Encoder Type:		
						Value	Meaning	
						0	Primary (differential) quad encoder.	
						1	No encoder (use motor back EMF for velocity estimation).	
						2	Analog encoder.	
						3	Secondary quad encoder from input lines.	
						4	Low frequency analog encoder.	
						5	Resolver.	
6	Use digital hall signals for position & velocity estimates.							

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						7	Analog encoder updated at current loop rate.
						8	Reserved for custom encoder.
						9	Panasonic
						10	SPI command (reserved for custom firmware use).
						11	EnDat
						12	SSI
						13	BiSS
						14	Serial encoders from Sanyo Denki, Tamagawa, Panasonic and HD systems.
						15	Custom encoders from HD systems.
						16	Simple analog potentiometer feedback.
						17	Reserved for custom encoder.
						18	Reserved for custom encoder.
						19	Reserved for custom encoder.
0x61	0x62	0x461	0x2383:21	F	INT16	Motor Encoder Units. This value defines the units used to describe linear motor encoders. It is not used with rotary motors.	
						Value	Description
						0	Microns.
						1	Nanometers.
						2	Millimeters.
0x62	0x63	0x462	0x2383:23	F	INT32	Motor Encoder Counts/Rev. Rotary motor only. When a resolver is used as the motor feedback, this parameter sets the resolution of the interpolated position. Units: counts/rev.	
0x63	0x64	0x463	0x2383:24	F	INT16	Motor Encoder Resolution. Linear motor only. Units: encoder units/count.	
0x64	0x65	0x464	0x2383:25	F	INT32	Motor Encoder Electrical Distance. Linear motor only. Units: encoder units/electrical cycle.	
0x65	0x66	0x465	0x2383:22	F	U16	Motor Encoder Direction. 0=normal, 1=reverse.	
0x67	0x68	0x467	0x2383:28	F	INT16	Analog Encoder Shift Amount. This value gives the number of bits of interpolation to be applied to an analog encoder. The fundamental encoder resolution will be increased by a multiplier of 2 ⁿ where n is the value programmed in this parameter. The range of this value is 0 to 10, giving possible multipliers of 1 to 1024.	

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0x68	0x69	0x468	0x2402	R*	INT32	Captured Index Position. Provides the position that the axis was in when an index pulse was captured. Configured by setting bits in the Position Capture Control Register (0x6C) , and the status of the captured data can be checked in the Position Capture Status Register (0x6D) . Reading this variable resets bits 0 & 3 of the Position Capture Status Register (0x6D) . Units: counts.	
0x69	0x6A	0x469	0x2232	R*	INT32	Unfiltered Motor Encoder Velocity. Units 0.1 counts/s.	
0x6A	0x6B	0x46A	0x2113	RF	INT32	Commanded Current Ramp Limit. Setting this to zero disables slope limiting. Units: mA/s.	
0x6B	0x6C	0x46B	0x2108	RF	9 or 14	Velocity Loop Command Filter Coefficients. A bi-quad filter structure that acts on the command input of the velocity loop just after velocity & acceleration limiting. 9 word parameter, see <i>Velocity Loop Filters</i> in the <i>CME 2 User Guide</i> . 14 word parameter (Plus product only), see Filter Coefficients (p. 73) .	
0x6C	0x6D	0x46C	0x2400	RF	INT16	Position Capture Control Register. Sets up position capture based on the index or home input. Bit-mapped as follows:	
						Bits	Description
						0	If set, the Captured Index Position (0x68) is captured on the falling edge of the index.
						1	If set, the Captured Index Position (0x68) is captured on the rising edge of the index.
						2	If set, a Captured Index Position (0x68) value will not be overwritten by a new position until it has been read. If clear, new positions will overwrite old positions.
						3, 4	Reserved.
						5	If set, Captured Home Position (0x10A) will be captured on the active to inactive edge of the home input switch. If clear, the home position will be captured on the inactive to active edge.
						6	If set, Captured Home Position (0x10A) will not be overwritten by a new position until it has been read. If clear, new positions will overwrite old positions.
						7	Reserved.
						8	If set, enable high speed input position capture..
						9	If set, don't overwrite high speed input capture positions.
						10	If set, latch high speed position capture.
12	Clear actual position on every encoder index pulse.						
0x6D	0x6E	0x46D	0x2401	R*	INT16	Position Capture Status Register. This register shows the current status of the index/home capture mechanism. Bit-mapped as follows:	

Parameter Dictionary	Parameters						
ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						Bits	Description
						0	If set, an index position has been captured. Cleared when the captured position is read.
						1-2	Reserved.
						3	If set, a new index transition occurred when a captured position was already stored. Depending on the mode, the new position may have been discarded, or overwritten the previously stored position.
						4	If set, a home position has been captured. This bit is cleared when the captured position is read.
						5-6	Reserved.
						7	If set, a new home pin transition occurred when a captured position was already stored. Depending on the mode, the new position may have been discarded, or overwritten the previously stored position.
						8	New high speed capture data if set.
						9-10	Reserved.
						11	High speed capture overflow if set.
0x6E	0x6F	0x46E	0x2383:34	F	INT16	Number of Resolver Cycles/Motor Rev. This parameter is only used with resolver feedback devices.	
0x6F	0x70	0x46F	0x2140	RF	INT16	PWM Mode and Status. Bit-mapped as follows:	
						Bits	Description
						0	Force bus clamping if set, disable bus clamping if clear. If bit 1 is set, then this bit is ignored.
						1	Automatic bus clamping mode if set. Setting this bit causes bus clamping mode to be automatically selected based on the output voltage. Bit 0 is ignored if this bit is set
						2	Factory reserved. If set, DBrk mode is enabled.
						4	Use hexagonal voltage limiting if set, circular limiting if clear. This setting is only used with brushless motors.
						6	1. Double PWM frequency if set.
0x70	0x71	0x470	0x2193:1	RF	See text	2. 8 3. Status bit, set when bus clamping is active.	
						Output 0 Configuration.	
						For notes on Output numbering see Input/Output Numbering .	
Data type is dependent on configuration and uses 1 to 5 words.							

Parameter Dictionary	Parameters																																	
ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description																												
						The first word is a bit-mapped configuration value. The remaining words give additional parameter data used by the output pin. Typically the second and third words are used as a 32-bit mask to identify which bit(s) in the status register the output should follow. If any of the selected bits in the status register are set, then the output will go active. If none of the selected bits are set, then the output will be inactive. Outputs may be programmed as a sync output for use in synchronizing multiple drives. In this configuration, the first word of this variable should be set to 0x0200 (i.e., only bit 9 is set), and the remaining words should be set to zero. Note that only output pin #0 has this feature. Attempting to program any other output pin as a sync output will have no effect. The first word is bit-mapped as follows: <table><tr><th>Bits</th><th>Configuration</th></tr><tr><td>0-4</td><td>Define which internal register drives the output. 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ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description		
							11	For EtherCAT drives, pulse on the SYNC0 signal.
							16	Track hardware position compare function on drives supporting it.
						5-7	Reserved for future use.	
						8	Inverts normal active state of output if set. E.g., outputs that are normally active low become active high.	
						9	If set, program the output as a sync output. This bit is reserved for all output pins except pin 0.	
						10-11	Reserved for future use.	
						12-13	Axis number for multi-axis drives.	
						14-15	Usage depends on output function selected.	
0x71	0x72	0x471	0x2193:2	RF	See text	Output 1 Configuration. See Output 0 Configuration (0x70) .		
0x72	0x73	0x472	0x 2193:3	RF	See text	Output 2 Configuration. See Output 0 Configuration (0x70) .		
0x73	0x74	0x473	0x 2193:4	RF	See text	Output 3 Configuration. See Output 0 Configuration (0x70) .		
0x74	0x75	0x474	0x 2193:5	RF	See text	Output 4 Configuration. See Output 0 Configuration (0x70) .		
0x75	0x76	0x475	0x 2193:6	RF	See text	Output 5 Configuration. See Output 0 Configuration (0x70) .		
0x76	0x77	0x476	0x 2193:7	RF	See text	Output 6 Configuration. See Output 0 Configuration (0x70) .		
0x77	0x78	0x477	0x 2193:8	RF	See text	Output 7 Configuration. See Output 0 Configuration (0x70) .		
0x78	0x79	0x478	0x 2192:1	RF	U16	Input 0 Configuration. Assigns a function to the input pin. All values not listed below are reserved for future use. For notes on Input numbering see Input/Output Numbering . The sync input function is only valid for high speed input pins. In addition, input pins 2 & 3 of the Accelus & Junus drives do not support this feature.		
						Bits	Configuration	
						0-7	Value	Meaning
							0	No function
							1	Reserved for future use (no function).
							2	Reset the drive on the rising edge of the input.
							3	Reset the drive on the falling edge of the input.
							4*	Positive side limit switch. Active high.
							5*	Positive side limit switch. Active low.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						6* Negative side limit switch. Active high.
						7* Negative side limit switch. Active low.
						8* Motor temperature switch. Active high.
						9* Motor temperature switch. Active low.
						10* Clear faults on rising edge, disable while high.
						11* Clear faults on falling edge, disable while low.
						12* Reset on rising edge. Disable drive while high.
						13* Reset on falling edge. Disable drive while low.
						14* Home switch. Active high.
						15* Home switch. Active low.
						16* Drive disable. Active high
						17* Drive disable. Active low.
						19 Sync input on falling edge.
						20* Halt motor, active high.
						21* Halt motor, active low.
						22 Scale analog input when high.
						23 Scale analog input when low.
						24* High speed position capture on rising edge. Only for high speed inputs.
						25* High speed position capture on falling edge. Only for high speed inputs.
						26 Count rising edges of input to indexer register (Register number identified by bits 8-11).
						27 Count falling edges of input to indexer register.
						28* Encoder fault input. Active high.
						29* Encoder fault input. Active low.
						30-35 Reserved.
						36* Abort move on rising edge if greater than n counts from destination position. Number of counts n is stored in an index register identified by bits 8-11.
						37* Abort move on falling edge if greater than n counts from destination position. Number of counts n is stored in an index register identified by bits 8-11.
						38* Mark HV loss on rising edge, disable while high.
						39* Mark HV loss on falling edge, disable while low.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						40* 41* 42* 43* 44* 45* *These functions use bit 8 to indicate that the input function should apply to all axes. This feature is only available in Plus products with a firmware version of at least 1.72 8-11 12-13
						Update trajectory on rising edge. Update trajectory on falling edge. Clear faults & event latch on rising edge. Clear faults & event latch on falling edge. Disable simulated encoder output when low. Burst current position on encoder output on rising edge. Disable simulated encoder output when high. Burst current position on encoder output on falling edge. Used to pass parameters to the input pin functions. Bits 12-13 are used to select the axis on multi-axis drives.
0x79	0x7A	0x479	0x2192:2	RF	U16	Input 1 Configuration. See Input 0 Configuration (0x78) .
0x7A	0x7B	0x47A	0x2192:3	RF	U16	Input 2 Configuration. See Input 0 Configuration (0x78) .
0x7B	0x7C	0x47B	0x2192:4	RF	U16	Input 3 Configuration. See Input 0 Configuration (0x78) .
0x7C	0x7D	0x47C	0x2192:5	RF	U16	Input 4 Configuration. See Input 0 Configuration (0x78) .
0x7D	0x7E	0x47D	0x2192:6	RF	U16	Input 5 Configuration. See Input 0 Configuration (0x78) .
0x7E	0x7F	0x47E	0x2192:7	RF	U16	Input 6 Configuration. See Input 0 Configuration (0x78) .
0x7F	0x80	0x47F	0x2192:8	RF	U16	Input 7 Configuration. See Input 0 Configuration (0x78) .
0x80	0x81	0x480	0x6503	F*	String	Model Number.
0x81	0x82	0x481	0x2384:1 or 0x1018:4	F*	U32	Drive Serial Number.
0x82	0x83	0x482	0x2384:3	F*	INT16	Drive's rated Peak Current. Units: 0.01 A.
0x83	0x84	0x483	0x2384:4	F*	INT16	Drive's rated Continuous Current. Units: 0.01 A.
0x84	0x85	0x484	0x2384:14	F*	INT16	Current Corresponding to Max A/D Reading. Units: 0.01 A.
0x85	0x86	0x485	0x2384:11	F*	U16	PWM Period. Units: 10 ns.
0x86	0x87	0x486	0x2384:12	F*	U16	Drive Servo Period (PWM periods). Servo loop update period as a multiple of the current loop period.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
0x87	0x88	0x487	None	F*	U16	Product Family. Identifies the drive product family. For specific drive hardware type, see Drive Hardware Type (0xAD) .	
0x88	0x89	0x488	0x2384: 5	F*	INT16	Drive's rated Time At Peak Current. The maximum time for which the drive is rated to output peak current. Units: ms.	
0x89	0x8A	0x489	0x2384:6	F*	INT16	Drive's rated Maximum Voltage. Maximum bus voltage rating. Units: 0.1 V.	
0x8A	0x8B	0x48A	0x2384:15	F*	INT16	Drive's rated Voltage Corresponding To Max A/D Reading. Units: 0.1 V.	
0x8B	0x8C	0x48B	0x2384:7	F*	INT16	Drive's rated Minimum Voltage. Minimum bus voltage rating. Units: 0.1 V.	
0x8C	0x8D	0x48C	0x2384:9	F*	INT16	Drive's rated Maximum Temperature. Units: degrees C.	
0x8D	0x8E	0x48D	0x2384:2	F*	String	Manufacturing info (date code, etc.).	
0x8E	0x8F	0x48E	0x2384:16	F*	INT16	Analog Input Scaling Factor. This is the voltage applied to the analog input which causes the max A/D value on the drive.	
0x90	0x91	0x490	None	R	U32	Serial Port Baud Rate. Units: bits/s. Defaults to 9600 at reset.	
0x91	0x92	0x491	None	R*	INT16	The maximum number of data words allowed per binary command over the serial interface.	
0x92	0x93	0x492	0x21A0	F	String	Drive Name. This object can assign an optional name to a drive. The data written here is stored to flash memory and is not used by the drive. Although this object is documented as holding a string (i.e. ASCII data), any values may be written here. Up to 40 bytes are stored.	
0x94	0x95	0x494	0x2384:24	R*	INT16	Firmware Version Number. The version number consists of a major and a minor version number. The minor number is passed in bits 0-7; the major number is in bits 8-15. For example, the version 1.12 would be encoded 0x010C.	
0x95	0x96	0x495	0x2421	F	20 words	Host Configuration State. Reserved for use by CME 2 software.	
0x96	0x97	0x496	0x2312	RF	INT16	Calibration Offset For Analog Reference. This voltage is added to the analog command input. It is factory-calibrated to give a zero reading for zero input voltage.	
0x97	0x98	0x497	0x2384:10	F*	INT16	Hysteresis Value For Drive Over Temperature Cut-Out. Units: degrees C.	
0x98	0x99	0x498	0x2330	RF	INT16	Function Generator Configuration. Configures the drive's internal function generator, which can drive the current, velocity, or position loop. Bit-mapped as follows:	
						Bits	Description
						0-1	Function code (type of waveform to generate):
							<table><tr><td>Value</td><td>Description</td></tr></table>
Value	Description						

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description								
							0	None (disabled).						
							1	Square wave.						
							2	Sine wave.						
						2-7	Reserved for future use.							
						8	If set, the function generator frequency is in units of 0.01 Hz. If clear, the frequency is in units of Hz.							
						9-11	Reserved.							
						12	One-shot mode if set. If bit 12 is set and bit 13 is clear, the function code is reset to zero (disabled) after one complete waveform. If bits 12 and 13 are both set, the function code is reset to zero after two waveforms.							
						13	Invert every other period if set.							
						14-15	Reserved for future use.							
						The function code programmed into bits 0-1 defines the type of waveform to be generated:								
						Code	Description							
						0	None (disabled).							
						1	Square wave.							
						2	Sine wave.							
						Note that the drive is placed under control of the function generator by setting Desired State (0x24) to one of the following values: 4 (function generator drives current loop); 14 (function generator drives velocity loop); 24 (function generator drives position loop in servo mode); 34 (function generator drives position loop in stepper mode).								
						Note that if one-shot mode is selected, then after one period (two if invert is selected) the function type will reset to zero.								
						0x99	0x9A	0x499	0x2331	RF	U16	Function Generator Frequency. Units: Hz. See bit 8 of Function Generator Configuration (0x98) .		
						0x9A	0x9B	0x49A	0x2332	RF	INT32	Function Generator Amplitude. The amplitude of the signal generated by the internal function generator. The units depend on the mode:		
												Mode	Units	
												Current	0.01 A.	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						Velocity 0.1 counts/s.
						Position Counts.
0x9B	0x9C	0x49B	0x2333	RF	U16	Function Generator Duty Cycle (square wave only). Units: 0.1% (for instance, 1000 for 100%).
0x9C	0x9D	0x49C	0x2384:8	F*	U16	Hysteresis For Maximum Bus Voltage Cut-Out. Units: 0.1 V.
0x9D	0x9E	0x49D	0x2384:18	F*	U16	PWM Dead Time At Continuous Current Limit. This parameter gives the PWM dead time used at or above the continuous current limit. The dead time below the continuous current limit is a linear function of this parameter and PWM Dead Time At Zero Current (0x9F) . Units: CPU cycles.
0x9E	0x9F	0x49E	0x2384:17	F*	U16	Drive Minimum PWM Off Time. This parameter gives the minimum amount of time for which all PWM outputs must be disabled for each current loop cycle. Units: 10 ns.
0x9F	0xA0	0x49F	0x2384:19	F*	U16	PWM Dead Time At Zero Current. This parameter gives the PWM dead time at zero current. The dead time above zero current is defined by a linear function of this parameter and PWM Dead Time At Continuous Current Limit (0x9D) . Units: CPU cycles.
0xA0	0xA1	0x4A0	0x1002	R*	U32	Drive Event Status Register. Bit-mapped as follows:
						Bits Description
						0 Short circuit detected.
						1 Drive over temperature.
						2 Over voltage.
						3 Under voltage.
						4 Motor temperature sensor active.
						5 Encoder feedback error.
						6 Motor phasing error.
						7 Current output limited.
						8 Voltage output limited.
						9 Positive limit switch active.
						10 Negative limit switch active.
						11 Enable input not active.
						12 Drive is disabled by software.
						13 Trying to stop motor.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						14 Motor brake activated.
						15 PWM outputs disabled.
						16 Positive software limit condition.
						17 Negative software limit condition.
						18 Tracking error.
						19 Tracking warning.
						20 Drive is currently in a reset condition.
						21 Position has wrapped. The Position variable cannot increase indefinitely. After reaching a certain value the variable rolls back. This type of counting is called position wrapping or modulo count.
						22 Drive fault. A drive fault that was configured as latching has occurred. For information on latching faults, see the CME 2 User Guide.
						23 Velocity limit has been reached.
						24 Acceleration limit has been reached.
						25 Tracking Window. Position Loop Error (0x35) is outside of Position Tracking Error Limit (0xBA).
						26 Home switch is active.
						27 In motion. Set if the trajectory generator is running a profile, or the Position Tracking Error Limit (0xBA) is outside the tracking window. Clear when settled into position.
						28 Velocity window. Set if the absolute velocity error exceeds the velocity window value.
						29 Phase not yet initialized. If the drive is phasing with no Halls, this bit is set until the drive has initialized its phase.
						30 Command fault. PWM or other command signal (e.g., EtherCAT master) not present. If Allow 100% Output option is enabled by a setting Bit 3 of Digital Input Command Configuration (0xA8) this fault will not detect a missing PWM command.
						31 Not defined.
0xA1	0xA2	0x4A1	0x2181	R	U32	Latched Event Status Register. This is a latched version of the Drive Event Status Register (0xA0). Bits are set by the drive when events occur. Bits are only cleared by writing to this parameter as explained below: When writing to the Latched Event Status Register, any bit set will cause the corresponding bit in the register to be cleared. For example, to clear the Over Voltage bit, write a 1 to the register. To clear all bits, write 0x4 to the register.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description																																								
0xA2	0xA3	0x4A2	0x2261	R*	INT16	Hall Input State. The lower three bits of the returned value give the present state of the Hall input pins. The Hall state is the value of the Hall lines AFTER the ordering and inversions specified in the Hall wiring configuration have been applied.																																								
0xA4	0xA5	0x4A4	0x2183	R	U32	Latching Fault Status Register. Bit-mapped to show which latching faults have occurred in the drive. When a latching fault has occurred, the fault bit (bit 22) of the Drive Event Status Register (0xA0) is set. The cause of the fault can be read from this register. To clear a fault condition, write a 1 to the associated bit in this register. The events that cause the drive to latch a fault are programmable. See Fault Mask (0xA7) for details. <table><tr><th colspan="2">Latched Faults</th></tr><tr><th>Bits</th><th>Fault Description</th></tr><tr><td>0</td><td>Data flash CRC failure. This fault is considered fatal and cannot be cleared.</td></tr><tr><td>1</td><td>Drive internal error. This fault is considered fatal and cannot be cleared.</td></tr><tr><td>2</td><td>Short circuit.</td></tr><tr><td>3</td><td>Drive over temperature.</td></tr><tr><td>4</td><td>Motor over temperature.</td></tr><tr><td>5</td><td>Over voltage.</td></tr><tr><td>6</td><td>Under voltage.</td></tr><tr><td>7</td><td>Feedback fault.</td></tr><tr><td>8</td><td>Phasing error.</td></tr><tr><td>9</td><td>Tracking error.</td></tr><tr><td>10</td><td>Current limited by I²T algorithm.</td></tr><tr><td>11</td><td>FPGA error type 1.</td></tr><tr><td>12</td><td>Command input lost.</td></tr><tr><td>13</td><td>FPGA error type 2.</td></tr><tr><td>14</td><td>Safety circuit fault.</td></tr><tr><td>15</td><td>Unable to control current.</td></tr><tr><td>16</td><td>Motor wiring disconnected (see Open Motor Wiring Check Current (0x19D)).</td></tr><tr><td>17-31</td><td>Reserved.</td></tr></table>	Latched Faults		Bits	Fault Description	0	Data flash CRC failure. This fault is considered fatal and cannot be cleared.	1	Drive internal error. This fault is considered fatal and cannot be cleared.	2	Short circuit.	3	Drive over temperature.	4	Motor over temperature.	5	Over voltage.	6	Under voltage.	7	Feedback fault.	8	Phasing error.	9	Tracking error.	10	Current limited by I ² T algorithm.	11	FPGA error type 1.	12	Command input lost.	13	FPGA error type 2.	14	Safety circuit fault.	15	Unable to control current.	16	Motor wiring disconnected (see Open Motor Wiring Check Current (0x19D)).	17-31	Reserved.
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17-31	Reserved.																																													
0xA5	0xA6	0x4A5	0x2191	RF	U16	Input Pin Configuration Register. Some drives have one or more pull-up resistors associated with their general-purpose input pins. On these drives, the state of the pull-ups can be controlled by writing to this register.																																								

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description																																		
						This register has one bit for each pull-up resistor available on the drive. Setting the bit causes the resistor to pull any inputs connected to it up to the high state when they are not connected. Bits 0 – 7 of this register are used to control pull-up resistor states. Each bit represents an input number. Bit 0 = IN1, bit 1 = IN2, etc. On drives that allow groups of inputs to be configured as either single ended or differential, bit 8 controls this feature. Set bit 8 to 0 for single ended, 1 for differential.																																		
0xA6	0xA7	0x4A6	0x2190	R*	U16	Input Pin States. The 16-bit value returned by this command gives the current state (high/low) of the drive’s input pins after debounce. Each bit represents one input as shown below. <table><tr><th>Bits</th><th>Description</th></tr><tr><td>0</td><td>Input 0.</td></tr><tr><td>1</td><td>Input 1.</td></tr><tr><td>2</td><td>Input 2.</td></tr><tr><td>3</td><td>Input 3.</td></tr><tr><td>4</td><td>Input 4.</td></tr><tr><td>5</td><td>Input 5.</td></tr><tr><td>6</td><td>Input 6.</td></tr><tr><td>7</td><td>Input 7.</td></tr><tr><td>8</td><td>Input 8.</td></tr><tr><td>9</td><td>Input 9.</td></tr><tr><td>10</td><td>Input 10.</td></tr><tr><td>11</td><td>Input 11.</td></tr><tr><td>12</td><td>Input 12.</td></tr><tr><td>13</td><td>Input 13.</td></tr><tr><td>14</td><td>Input 14.</td></tr><tr><td>15</td><td>Input 15.</td></tr></table>	Bits	Description	0	Input 0.	1	Input 1.	2	Input 2.	3	Input 3.	4	Input 4.	5	Input 5.	6	Input 6.	7	Input 7.	8	Input 8.	9	Input 9.	10	Input 10.	11	Input 11.	12	Input 12.	13	Input 13.	14	Input 14.	15	Input 15.
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14	Input 14.																																							
15	Input 15.																																							

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description																						
0xA7	0xA8	0x4A7	0x2182	RF	U32	Fault Mask. This variable is used to configure which drive events cause latching faults. Setting a fault mask bit to 1 causes the associated drive event to cause a latching fault when it occurs. Setting a fault mask bit to 0 disables fault latching on the associated event. Latched faults may be cleared using the Latching Fault Status Register (0xA4). <table><tr><th>Bits</th><th>Fault Description</th></tr><tr><td>0</td><td>Data flash CRC failure. This bit is read-only and will always be set. If the drive detects corrupted flash data values on startup it will remain disabled and indicate a fault condition.</td></tr><tr><td>1</td><td>Drive internal error. This bit is read-only and will always be set. If the drive fails its power-on self-test, it will remain disabled and indicate a fault condition.</td></tr><tr><td>2</td><td>Short circuit. If set: programs the drive to latch a fault condition when a short circuit is detected on the motor outputs. If clear: programs the drive to disable its outputs for 100 ms after a short circuit and then re-enable.</td></tr><tr><td>3</td><td>Drive over temperature. If set: programs the drive to latch a fault condition when a drive over temperature event happens. If clear: programs the drive to re-enable as soon as it cools sufficiently from an over temperature event.</td></tr><tr><td>4</td><td>Motor over temperature. If set: programs the drive to latch a fault condition when a motor temperature sensor input activates. If clear: programs the drive to re-enable as soon as the over temperature input becomes inactive.</td></tr><tr><td>5</td><td>Over voltage. If set: programs the drive to latch a fault condition when excessive bus voltage is detected. If clear: programs the drive to re-enable as soon as the bus voltage is within normal range.</td></tr><tr><td>6</td><td>Under voltage. If set: programs the drive to latch a fault condition when inadequate bus voltage is detected. If clear: programs the drive to re-enable as soon as the bus voltage is within normal range.</td></tr><tr><td>7</td><td>Feedback fault. If set: programs the drive to latch a fault when feedback faults occur. Feedback faults occur if too much current is drawn from the 5 V source on the drive, a resolver or analog encoder is disconnected, or a resolver or analog encoder has levels out of tolerance.</td></tr><tr><td>8</td><td>Phasing error. If set: programs the drive to latch a fault when phasing errors occur. If clear: programs the drive to re-enable when the phasing error is removed.</td></tr><tr><td>9</td><td>Tracking error. If set: programs the drive to latch in the disabled state when a tracking error</td></tr></table>	Bits	Fault Description	0	Data flash CRC failure. This bit is read-only and will always be set. If the drive detects corrupted flash data values on startup it will remain disabled and indicate a fault condition.	1	Drive internal error. This bit is read-only and will always be set. If the drive fails its power-on self-test, it will remain disabled and indicate a fault condition.	2	Short circuit. If set: programs the drive to latch a fault condition when a short circuit is detected on the motor outputs. If clear: programs the drive to disable its outputs for 100 ms after a short circuit and then re-enable.	3	Drive over temperature. If set: programs the drive to latch a fault condition when a drive over temperature event happens. If clear: programs the drive to re-enable as soon as it cools sufficiently from an over temperature event.	4	Motor over temperature. If set: programs the drive to latch a fault condition when a motor temperature sensor input activates. If clear: programs the drive to re-enable as soon as the over temperature input becomes inactive.	5	Over voltage. If set: programs the drive to latch a fault condition when excessive bus voltage is detected. If clear: programs the drive to re-enable as soon as the bus voltage is within normal range.	6	Under voltage. If set: programs the drive to latch a fault condition when inadequate bus voltage is detected. If clear: programs the drive to re-enable as soon as the bus voltage is within normal range.	7	Feedback fault. If set: programs the drive to latch a fault when feedback faults occur. Feedback faults occur if too much current is drawn from the 5 V source on the drive, a resolver or analog encoder is disconnected, or a resolver or analog encoder has levels out of tolerance.	8	Phasing error. If set: programs the drive to latch a fault when phasing errors occur. If clear: programs the drive to re-enable when the phasing error is removed.	9	Tracking error. If set: programs the drive to latch in the disabled state when a tracking error
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ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						occurs. If clear: programs the drive to abort the current move and remain enabled when a tracking error occurs. 10 If set: programs the drive to latch a fault when output current is limited by the I²T algorithm. 11 FPGA failure. This bit is read-only. 12 Command input lost fault. If set: programs the drive to latch in the disabled state when the command input is lost. 13 Unable to initialize internal drive hardware. This bit is read-only. 14 If set, programs the drive to latch a fault when there is safety circuit consistency check failure. 15 If set, programs the drive to latch a fault when the drive is unable to control motor current. 16 If set, programs the drive to latch a fault when the motor wiring is disconnected (see Open Motor Wiring Check Current (0x19D)). 17-31 Reserved.
0xA8	0xA9	0x4A8	0x2320	RF	INT16	Digital Input Command Configuration. Defines the configuration of the digital input commands when the drive is running in a mode that uses them as a control source. The lower 8 bits control the PWM input configuration for controlling current and velocity modes. The upper 8 bits configure the digital inputs when running in position mode. Bits Description 0 If set, use PWM in signed/magnitude mode. If clear, use PWM in 50% duty cycle offset mode. 1 Invert the PWM input if set. 2 Invert the sign input if set. 3 Allow 100% duty cycle if set. If clear, treat 100% duty cycle as a zero command, providing a measure of safety in case of controller failure or cable break. 4 If set, use parameter 0xB6 as deadband for PWM input. 5 If set, allow longer PWM periods (up to 50ms). 6 For Feature Set C products, setting this bit will cause the Motor Hall Offset (0x4F) to be added to the angle calculated in UV mode. For Feature Set E products, see UV Configuration (0x180). 7 Reserved for future use. 8-9 Input pin interpretation for position mode (see below).

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description		
							Value	Description
							0	Step & Direction mode.
							1	Separate up & down counters.
							2	Quadrature encoder input.
							3	PWM input commands absolute position.
						10-11	Reserved for future use.	
						12	Pulses are counted on their falling edge if this bit is clear, rising edge if set. This bit has no effect when the inputs are configured as encoder inputs.	
						13	Causes the direction of the input to be reversed. Works for all three modes..	
						14-15	Identify which input pins to use.	
							Value	Description
							0	Single ended high speed inputs.
							1	Multi-mode encoder port.
							2	Differential high speed inputs.
						3	Use primary encoder inputs.	
0xA9	0xAA	0x4A9	0x2321	RF	INT32	Digital Command Input Scaling Factor. This value gives the amount of current to command at 100% PWM input. The scaling depends on what the PWM input is driving: Current mode: 0.01 A Velocity (Accelus): 0.1 counts/s In position mode the scaling factor is a ratio of two 16-bit values. The first word passed gives the numerator and the second word gives the denominator. This ratio determines the number of encoder units moved for each pulse (or encoder count) input. For example, a ratio of 1/3 would cause the motor to move 1 encoder unit for every three input steps.		
0xAA	0xAB	0x4AA	0x2196	R*	U16	Raw Input State. The 16-bit value returned by this command gives the current state (high/low) of the drive's input pins. Unlike Input Pin States (0xA6) , no debounce is applied when reading the inputs using this variable. The bits are mapped in the same order as Input Pin States (0xA6) .		
0xAB	0xAC	0x4AB	0x2194	R	U16	Output States And Program Control. When read, this parameter gives the		

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description																														
						active/inactive state of the drive's general-purpose digital outputs. Each bit represents an input number. Bit 0 = digital output 0 (OUT0), bit 1 = output 1 (OUT1), etc., up to output n (OUTn), the number of digital outputs on the drive. Additional bits are ignored. Outputs that have been configured for program control can be set by writing to this parameter. Set a bit to activate the output. It will be activated high or low according to how it was programmed. Clear a bit to make the output inactive. If an output was not configured for program control it will not be affected.																														
0xAC	0xAD	0x4AC	0x2180	R*	U32	Sticky Drive Event Status Register. This read-only parameter is bit-mapped in exactly the same way as the Drive Event Status Register (0xA0), but instead of giving the present status of the drive, the sticky version indicates any bits in the event status that have been set since the last reading of the sticky register. The sticky register is similar to the Latched Event Status Register (0xA1), but the latched register must be cleared explicitly, whereas the sticky register is cleared automatically each time it is read.																														
0xAD	0xAE	0x4AD	0x1018:2 or 0x2384:13	R*	INT16	Drive Hardware Type. Also known as Product Code. Identifies the specific drive model. This is an augmented version of Product Family (0x87). <table><tr><th>Value</th><th>Product</th></tr><tr><td>0x0000</td><td>ASC: Accelus Card.</td></tr><tr><td>0x0002</td><td>ASP: Accelus Panel.</td></tr><tr><td>0x0100</td><td>JSP: Junus Panel.</td></tr><tr><td>0x0200</td><td>ACM: Accelnet Module.</td></tr><tr><td>0x0201</td><td>XSL: Xenus Panel (obsolete).</td></tr><tr><td>0x0203</td><td>ACP: Accelnet Panel (obsolete).</td></tr><tr><td>0x0206</td><td>XSL-R: Xenus Panel, resolver version.</td></tr><tr><td>0x0207</td><td>XSL: Xenus Panel.</td></tr><tr><td>0x0209</td><td>ACJ: Accelnet Micro Panel.</td></tr><tr><td>0x020B</td><td>ACP: Accelnet Panel.</td></tr><tr><td>0x020C</td><td>ACK: Accelnet Micro Module.</td></tr><tr><td>0x020E</td><td>Special.</td></tr><tr><td>0x020F</td><td>Special.</td></tr><tr><td>0x0210</td><td>ACJ-S: Accelnet Micro Panel, analog encoder version.</td></tr></table>	Value	Product	0x0000	ASC: Accelus Card.	0x0002	ASP: Accelus Panel.	0x0100	JSP: Junus Panel.	0x0200	ACM: Accelnet Module.	0x0201	XSL: Xenus Panel (obsolete).	0x0203	ACP: Accelnet Panel (obsolete).	0x0206	XSL-R: Xenus Panel, resolver version.	0x0207	XSL: Xenus Panel.	0x0209	ACJ: Accelnet Micro Panel.	0x020B	ACP: Accelnet Panel.	0x020C	ACK: Accelnet Micro Module.	0x020E	Special.	0x020F	Special.	0x0210	ACJ-S: Accelnet Micro Panel, analog encoder version.
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ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						0x0240 STM: Stepnet Module.
						0x0242 STP: Stepnet Panel.
						0x0243 STL: Stepnet Micro Module.
						0x0300 ASP: Accelnet Panel, dual axis.
						0x0310 XSJ(S): Xenus Micro Panel.
						0x0320 XTL: Xenus Panel, resolver version.
						0x0330 XTL(S): Xenus Panel.
						0x0340 XSJ-R: Xenus Micro Panel, resolver version.
						0x0380 AEP: Accelnet EtherCAT Panel.
						0x0391 AMP: Accelnet Macro Panel.
						0x0350 STX: Stepnet AC Panel.
						0x0370 ACK-R: Accelnet Micro Module, resolver version.
						0x03A0 ADP: Accelnet Panel.
						0x1000 XEL: Xenus Plus EtherCAT.
						0x1010 XML: Xenus Plus MACRO.
						0x1020 XPL: Xenus Plus CAN.
						0x1030 AEM: 1 axis Accelnet Plus Module EtherCAT.
						0x1040 APM: 1 axis Accelnet Plus Module CAN.
						0x1050 AE2: 2 axis Accelnet Plus Module EtherCAT.
						0x1060 AP2: 2 axis Accelnet Plus Module CAN.
						0x1070 SEM: 1 axis Stepnet Plus Module EtherCAT.
						0x1080 SPM: 1 axis Stepnet Plus Module CAN.
						0x1090 SE2: 2 axis Stepnet Plus Module EtherCAT.
						0x10A0 SP2: 2 axis Stepnet Plus Module CAN.
						0x10B0 XE2: 2 axis Xenus Plus Dual EtherCAT
						0x10B8 XE2-R: 2 axis Xenus Plus Dual EtherCAT resolver version
						0x10C0 BE2: 2 axis Accelnet Plus Panel EtherCAT
						0x10C8 BE2-R: 2 axis Accelnet Plus Panel EtherCAT resolver version
						0x10D0 XP2: 2 axis Xenus Plus Dual CAN
						0x10D8 XP2-R: 2 axis Xenus Plus Dual CAN resolver version
						0x10E0 BP2: 2 axis Accelnet Plus Panel CAN

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						0x10E8	BP2-R: 2 axis Accelnet Plus Panel CAN resolver version
						0x10F0	TE2: 2 axis Stepnet Plus Panel EtherCAT
						0x1100	TP2: 2 axis Stepnet Plus Panel CAN
						0x1110	BEL: 1 axis Accelnet Plus Panel EtherCAT
						0x1118	BEL-R: 1 axis Accelnet Plus Panel EtherCAT resolver version
						0x1120	BPL: 1 axis Accelnet Plus Panel CAN
						0x1128	BPL-R: 1 axis Accelnet Plus Panel CAN resolver version
						0x1130	TEL: 1 axis Stepnet Plus Panel EtherCAT
						0x1150	SP4: 4 axis Stepnet Plus Module CAN
0xAE	0xAF	0x4AE	0x60F6:3	RF	INT16	Current Loop Offset. This value is added to the commanded motor current. It can compensate for a directional bias affecting the current loop. Units: 0.01 A.	
0xAF	0xB0	0x4AF	0x2420	RF	INT32	Miscellaneous Drive Options Register. Bit-mapped as follows:	
						Bits	Option
						0	If set, input pins 1, 2, and 3 are pulled high on the drive. If clear, the pins are not pulled up. This option is only available on the Junus drive.
						1	Reserved.
						2	If set, limit switch inputs will only abort a trajectory in progress, but will not affect current output. If clear, limit switches limit current.
						3	If set, save PDO configuration to a file in the CVM file system when a "Save to Flash" command is received over the CANopen network. If clear, a PDO is not saved.
						4	If set, a limit switch activation will be treated as a fault in the CANopen Status Word (CANopen index 0x6041 as described in the <i>CANopen Programmer's Manual</i>).
						5-31	Reserved.
0xB0	0xB1	0x4B0	0x2260	R	INT16	Motor Phase Angle. Writes are only useful when running in diagnostic micro-stepping mode. Units: degrees.	
0xB1	0xB2	0x4B1	0x21C1	RF	INT16	Increment Rate For Phase Angle When In Micro Stepping Mode. Only used in diagnostic micro-stepping mode. Desired State (0x24) = 42 (microstepping mode). Units: degrees/s.	
0xB2	0xB3	0x4B2	0x21C0	RF	U16	Commutation Mode. Also known as Phasing Mode. Controls the mechanism used by the drive to compute the motor phasing angle. Determines what inputs the drive uses to initialize and maintain the phase angle, as follows:	
						Value	Mode

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						0 Standard mode. Encoder-based sinusoidal commutation for brushless motors. Use digital Hall inputs to initialize phase, then switch to an encoder to maintain phase. The encoder is the primary sensing device with the Hall effect sensors used to monitor and adjust the phase angle as necessary during operation.
						1 Trapezoidal (Hall based) phasing. The Hall sensors are used for phasing all the time. This mode can be used if no encoder is available.
						2 Like mode 0 except that the phase angle is not adjusted based on the Hall inputs. Hall sensors are still required to initialize the phase angle at startup.
						3 Analog Halls (90 degrees). Only available on drives with the necessary analog Hall inputs.
						4 DC brush motor mode.
						5 Algorithmic Phase Initialization mode (wake & wiggle, no Halls). See the <i>CME 2 User Guide</i> for more information on Algorithmic Phase Initialization.
						6 Encoder based phasing. Use with resolver or Servo-Tube motors.
						7 Trapezoidal commutation with phase angle interpolation.
0xB3	0xB4	0x4B3	0x2384:23	F*	INT16	Analog Encoder Scaling Factor. This parameter selects the resolution of an analog encoder input. The parameter is not used for other encoder types.
0xB4	0xB5	0x4B4	0x2263	R*	INT16	Encoder Phase Angle. For feedback types, such as resolver, that can also calculate phase angle information. This parameter allows the phase information to be read directly.
0xB5	0xB6	0x4B5	0x2353	R*	INT32	Homing Adjustment. This parameter is updated after each successful homing operation. The value it contains is the size of the actual position adjustment made in the last home sequence. Units: counts.
0xB6	0xB7	0x4B6	0x2322	RF	U16	PWM Input Frequency. This is the frequency of the PWM for use in UV commutation mode only. Units: 10 Hz. This parameter is also used to specify an optional PWM dead band when running in normal (not UV) PWM command modes. When used as a dead band value, this input should be set in the range 0 to 32767 which corresponds to a dead band of 0 to 100% of the PWM duty cycle.
0xB7	0xB8	0x4B7	0x2141	R*	U32	System Time. Time since start up. Units: ms.
0xB8	0xB9	0x4B8	0x607D:2	RF	INT32	Positive Software Limit. This parameter is only available on drives that support trajectory generation and homing. Software limits are only in effect after the drive has been referenced (i.e. homing has been successfully completed). Set to less than negative software limit to disable. Units: counts.
0xB9	0xBA	0x4B9	0x607D:1	RF	INT32	Negative Software Limit. Software limits are only in effect after the drive has been

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description		
						referenced (i.e. homing has been successfully completed). Set to greater than positive software limit to disable. Units: counts.		
0xBA	0xBB	0x4BA	0x2120	RF	INT32	Position Tracking Error Limit. Also known as Tracking Error Window. If the Position Loop Error (0x35) exceeds this value then the tracking error bit (bit 18) of the Drive Event Status Register (0xA0) is set and the motor is stopped. Using Fault Mask (0xA7) , the tracking error event can be configured to either disable the drive immediately, or abort the present move and continue holding position. Units: counts.		
0xBB	0xBC	0x4BB	0x6065	RF	INT32	Position Tracking Warning Limit. If the Position Loop Error (0x35) exceeds this value then the tracking warning bit (bit 19) of the Drive Event Status Register (0xA0) is set. Units: counts.		
0xBC	0xBD	0x4BC	0x6067	RF	INT32	Position Tracking Window Limit. If the Position Loop Error (0x35) exceeds this value then the tracking window bit (bit 25) of the Drive Event Status Register (0xA0) is set. Units: counts.		
0xBD	0xBE	0x4BD	0x6068	RF	U16	Time Delay For Position Tracking Error Limit (0xBA) . The tracking window bit (bit 25) of the Drive Event Status Register (0xA0) will not be cleared until the Position Loop Error (0x35) has been within the Position Tracking Error Limit (0xBA) for at least this long. Units: ms.		
0xBE	0xBF	0x4BE	0x2253	RF	U32	Software Limit Deceleration. The deceleration rate used to stop a motor when approaching a software limit. Units: 10 counts/s ² .		
0xBF	0xC0	0x4BF	0x2351	RF	U16	Homing Current Delay Time (used with home to hard stop mode only). When performing a home to hard stop, the amplifier will push against the stop for this long before sampling the home position. Units: ms.		
0xC0	0xC1	0x4C0	None	R*	INT16	Network Node ID. This is the drive's present ID as read at system startup. The node ID is only read at system startup, so this value will not change unless the drive is reset. See Network Node ID Configuration (0xC1) .		
0xC1	0xC2	0x4C1	0x21B0	RF	INT16	Network Node ID Configuration. Defines how a drive's node ID is calculated, and specifies the drive's network bit rate. The ID is calculated at startup (and only at startup) using a combination of general-purpose input pins and a programmed offset value. On certain models, an address switch is also used. The resulting value is clipped to a 7-bit ID in the range 0 to 127. Bit-mapped as follows:		
						<table><tr><th>Bits</th><th>Description</th></tr></table>	Bits	Description
Bits	Description							

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						0-6	Give the node ID offset value that will be added to the value read from the input pins and address selector switch.
						7	Used only on DeviceNet firmware. If this bit is set, then the drive will be software disabled on startup and will remain disabled until it is enabled by a DeviceNet I/O message with the enable bit set.
						8-10	Number of input pins (0-7) to read on startup for the node ID value. If input pins are used (i.e., the value in bits 8-10 is not zero), the inputs can be mapped to node ID bits through the parameter Input Pin Mapping For Node ID Selection (0x103) .
						11	This bit is ignored on drives that do not have an address switch. On drives with an address switch, setting this bit programs the drive to use the address selector switch as part of the address calculation. In this case, the node ID value is equal to the sum of: ▪ The value read from the designated input pins, shifted up 4 bits. ▪ The address switch value. ▪ The programmed offset value. Note that since the node ID is always clipped to the lowest 7 bits, no more than 3 input pins will ever have an effect on the node address when the address switch is used.
						12-15	Network bit rate setting:
							Value Bit Rate (bits/s)
							0 1,000,000.
							1 800,000.
							2 500,000.
							3 250,000.
							4 125,000.
							5 50,000.
							6 20,000.
							7-15 Reserved for future use
0xC2	0xC3	0x4C2	0x2352	RF	INT16	Homing Method Configuration. Bit-mapped as follows:	
						Bits	Description
						0-3	Home function.
							Value Description
						0	If bit 5 is not set, then just set the current position as home. If bit 5 is set, then move in the direction specified by bit 4 and set the location of the first index pulse as

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						home. Bit 6 is not used in this mode.
					1	Move in the direction specified by bit 4 until a limit switch is encountered. Then move in the other direction out of limit. If bit 5 is clear, then the edge location is home. If bit 5 is set, then the next index pulse is home. Bit 6 not used in this mode.
					2	Home on a constant home switch. The initial move is made in the direction specified by bit 4. When the home switch is encountered, the direction is reversed. The edge of the home switch is set as home if bit 5 is clear. If bit 5 is set, then an index pulse is used as the home position. Bit 6 is used to define which index pulse is used.
					3	Home on an intermittent home switch. This mode works the same as mode 2 except that if a limit switch is encountered when initially searching for home, then the direction is reversed. In mode 2, hitting a limit switch before finding home would be considered an error. Bit 8 identifies which edge of the home to search for (positive or negative).
					4	Home to a hard stop. This moves in the direction specified in bit 4 until the home current limit is reached. It then presses against the hard stop using that current value until the home delay time expires. If bit 5 (index) is set, drive away from the hard stop until an index is found.
					15	Immediate home. This value causes the amp to be referenced immediately on power-up. Once the encoder is initialized, the home offset value is added to the encoder position and the result is set as the current referenced position. This is primarily useful with absolute encoders.
					4	Initial move direction (0=positive, 1=negative).
					5	Home on index pulse if set.
					6	Selects which index pulse to use. If set, use the pulse on the DIR side of the sensor edge. DIR is the direction specified by bit 4 of this word.
					7	If set, capture falling edge of index. Capture rising edge if clear.
					8	When using a momentary home switch, this bit identifies which edge of the home switch to reference on. If set, then the negative edge is used; if clear the positive edge is used.
					9	If set, make a move to the zero position when homing is finished. If clear, the zero position is found, but not moved to.
					10	If set, the homing sequence will run as normal, but the actual position will not be adjusted at the end. Note that even though the actual position is not adjusted, the parameter Homing Adjustment (0xB5) is updated with the size of the adjustment (in counts) that would have been made. Also, if bit 10 is set then no move to zero is made regardless of the setting of bit 9.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description		
0xC3	0xC4	0x4C3	0x6099:1	RF	INT32	Homing Velocity (fast moves). This velocity value is used during segments of the homing procedure that may be handled at high speed. Generally, this means moves in which the home sensor is being located, but the edge of the sensor is not being found. Units: 0.1 counts/s.		
0xC4	0xC5	0x4C4	0x6099:2	RF	INT32	Homing Velocity (slow moves). This velocity value is used for homing segments that require low speed, such as cases where the edge of a homing sensor is being sought. Units: 0.1 counts/s.		
0xC5	0xC6	0x4C5	0x609A	RF	U32	Homing Acceleration/Deceleration. This value defines the acceleration used for all homing moves. The same value is used at the beginning and ending of moves (i.e. there is no separate deceleration value). Units: 10 counts/s ² .		
0xC6	0xC7	0x4C6	0x607C	RF	INT32	Home Offset. The home offset is the difference between the zero position for the application and the machine home position (found during homing). During homing the home position is found. Once the homing is completed the zero position is offset from the home position by adding the offset to the home position. All subsequent absolute moves shall be taken relative to this new zero position. Units: counts.		
0xC7	0xC8	0x4C7	0x2350	RF	INT16	Homing Current Limit (home to hard stop mode only). Home current in hard stop mode, in which the drive drives the motor to the mechanical end of travel (hard stop). End of travel is recognized when the drive outputs the Hard Stop Mode Home Current for the Homing Current Delay Time (0xBF) .Units: 0.01 A.		
0xC8	0xC9	0x4C8	None	RF	INT16	Trajectory Profile Mode. To set profile in CANopen see CAN object 0x6086 in the CANopen Programmers Manual. Bit-mapped as follows:		
						Bits	Description	
						0-2	Give the trajectory profile mode. The possible trajectory modes are described below.	
							Value	Description
							0	Trapezoidal profile mode. Uses position/distance, velocity, acceleration and deceleration. Any of those parameters may be changed during the move. Jerk is not used in this mode.
							1	S-curve profile mode. Uses position/distance, velocity, acceleration, and jerk. None of these parameters may be changed while a move is in progress (although the move may be aborted). The acceleration parameter will be used for deceleration.
2	Velocity mode. Uses velocity, acceleration, and deceleration. Jerk is not used in this mode, and position is only used to define the direction of move (zero or positive to move with a positive velocity, negative to move with a negative velocity). Any parameter may be changed during the move. Set velocity to zero to stop.							

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description		
							3	PVT profile mode. Use of this mode through the serial interface is not presently supported.
						8	If set, this is a relative move. If clear, this is an absolute move.	
						Valid settings for this parameter:		
						Value	Description	
						0	Absolute move, trapezoidal profile.	
						1	Absolute move, S-curve profile.	
						256	Relative move, trapezoidal profile.	
						257	Relative move, S-curve profile.	
						2	Velocity profile.	
0xC9	0xCA	0x4C9	0x2252	R*	INT16	Trajectory Status Register. This parameter gives status information about the trajectory generator. Bit-mapped as follows:		
						Bits	Description	
						0-8	Reserved for future use.	
						9	Cam table underflow.	
						10	Reserved for future use.	
						11	Homing error. If set, an error occurred in the last home attempt. Cleared by a home command.	
						12	Referenced. Set when a homing command has been successfully executed. Cleared by a home command.	
						13	Homing. If set, the drive is running a home command.	
						14	Set when a move is aborted. Cleared at the start of the next move.	
15	In-Motion Bit. If set, the trajectory generator is presently generating a profile.							
0xCA	0xCB	0x4CA	0x607A	RF	INT32	Trajectory Generator Position Command. Units: Counts. Meaning depends on move type as described below:		
						Move Type		Meaning
						Relative		Move distance.
						Absolute		Target position.
						Velocity		Direction: 1 for positive, -1 for negative.
0xCB	0xCC	0x4CB	0x6081	RF	INT32	Trajectory Maximum Velocity. The trajectory generator will attempt to reach this velocity during a move. Units: 0.1 counts/s.		

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
0xCC	0xCD	0x4CC	0x6083	RF	U32	Trajectory Maximum Acceleration. The trajectory generator will attempt to reach this acceleration during a move. For s-curve profiles, this value is also used to decelerate at the end of a move. Units: 10 counts/s ² .
0xCD	0xCE	0x4CD	0x6084	RF	U32	Trajectory Maximum Deceleration. In trapezoidal trajectory mode, this value will be used to decelerate at the end of a move. Units: 10 counts/s ² .
0xCE	0xCF	0x4CE	0x2121	RF	U32	Trajectory Maximum Jerk. Also known as Trajectory Jerk Limit. The S-curve profile generator uses this value as the jerk (rate of change of acceleration/deceleration) during moves. Other profiles types do not use jerk limit. Units: 100 counts/s ³ .
0xCF	0xD0	0x4CF	0x6085	RF	U32	Trajectory Abort Deceleration. If a move is aborted, this value will be used by the trajectory generator to decelerate to a stop. Units: 10 counts/s ² .
0xD0	0xD1	0x4D0	0x2192:9	RF	U16	Input 8 Configuration. See Input 0 Configuration (0x78) .
0xD1	0xD2	0x4D1	0x2192:10	RF	U16	Input 9 Configuration. See Input 0 Configuration (0x78) .
0xD2	0xD3	0x4D2	0x2192:11	RF	U16	Input 10 Configuration. See Input 0 Configuration (0x78) .
0xD3	0xD4	0x4D3	0x2192:12	RF	U16	Input 11 Configuration. See Input 0 Configuration (0x78) .
0xD4	0xD5	0x4D4	0x2192:13	RF	U16	Input 12 Configuration. See Input 0 Configuration (0x78) .
0xD5	0xD6	0x4D5	0x2192:14	RF	U16	Input 13 Configuration. See Input 0 Configuration (0x78) .
0xD6	0xD7	0x4D6	0x2192:15	RF	U16	Input 14 Configuration. See Input 0 Configuration (0x78) .
0xD7	0xD8	0x4D7	0x2192:16	RF	U16	Input 15 Configuration. See Input 0 Configuration (0x78) .
0xD8	0xD9	0x4D8	0x2150	RF	U16	Regen Resistor Resistance. Units: 0.1 Ω.
0xD9	0xDA	0x4D9	0x2151	RF	U16	Regen Resistor, Continuous Power. Units: W.
0xDA	0xDB	0x4DA	0x2152	RF	U16	Regen Resistor, Peak Power. Units: W.
0xDB	0xDC	0x4DB	0x2153	RF	U16	Regen Resistor, Time At Peak. Units: ms.
0xDC	0xDD	0x4DC	0x2154	RF	INT16	Regen Turn On Voltage Units: 0.1 V.
0xDD	0xDE	0x4DD	0x2155	RF	INT16	Regen Turn Off Voltage. Units: 0.1 V.
0xDE	0xDF	0x4DE	0x2384:20	F*	INT16	Drive's Peak Current Rating For Its Internal Regen Transistor. Units: 0.01 A.
0xDF	0xE0	0x4DF	0x2384:21	F*	INT16	Drive's Continuous Current Rating For Its Internal Regen Transistor.
0xE0	0xE1	0x4E0	0x2384:22	F*	INT16	Drive's Time At Peak Current For Its Internal Regen Transistor. Units: ms.
0xE1	0xE2	0x4E1	0x2156	F	String	Regen Resistor Model Number String.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
0xE2	0xE3	0x4E2	0x2157	R*	INT16	Regen Resistor Status. Bit-mapped as follows:	
						Bits	Description
						0	Set if the regen circuit is currently closed.
						1	Set if regen is required based on bus voltage.
						2	Set if the regen circuit is open due to an overload condition. The overload may be caused by either the resistor settings or the internal drive protections.
						3-15	Reserved.
0xE3	0xE4	0x4E3	0x2382	RF	U16	Position Loop Output Gain Multiplier. The output of the position loop is multiplied by this value before being passed to the velocity loop. This scaling factor is calculated such that a value of 100 is a 1.0 scaling factor. This parameter is most useful in dual loop systems.	
0xE4	0xE5	0x4E4	0x21C2	RF	INT16	Maximum Current to use with algorithmic phase initialization. See code 5 of Commutation Mode (0xB2) . Units: 0.01 A.	
0xE5	0xE6	0x4E5	0x21C3	RF	U16	Algorithmic Phase Initialization Timeout. See code 5 of Commutation Mode (0xB2) . Units: ms.	
0xE6	0xE7	0x4E6	0x21D8	RF	INT32	Maximum Velocity Adjustment. This is the maximum velocity adjustment made by the stepper outer position loop when enabled. This parameter is only used when the stepper outer loop is engaged (when bit 1 of Stepper Configuration & Status (0xEE) is set). Units: 0.1 steps/s.	
0xE7	0xE8	0x4E7	0x21D7	RF	U16	Proportional Gain For Stepper Outer Loop. This parameter gives the gain used for calculating a velocity adjustment based on Position Loop Error (0x35) . This parameter is only used when the stepper outer loop is engaged (when bit 1 of Stepper Configuration & Status (0xEE) is set).	
0xE8	0xE9	0x4E8	0x21D0	RF	INT16	Holding Current For Microstepping Mode. Units: 0.01 A.	
0xE9	0xEA	0x4E9	0x21D1	RF	U16	Run to Hold Time For Microstepping Mode. Units: ms.	
0xEA	0xEB	0x4EA	0x21D2	RF	U16	Detent Correction Gain Factor For Microstepping Mode.	
0xED	0xEE	0x4ED	0x21D5	RF	U16	Holding Current To Fixed Voltage Output Time for Microstepping Mode. Time delay from entering hold current before entering the special voltage control mode of operation. This mode trades the normal tight control of current for very low jitter on the motor position. Used in stepper mode only. Set to 0 to disable this feature. Units: ms.	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
0xEE	0xEF	0x4EE	0x21D6	RF	INT16	Stepper Configuration & Status. Bit-mapped as follows:	
						Bits	Description
						0	Use the encoder input for phase compensation if enabled. Pure stepper mode if disabled.
						1	Use on outer position loop to adjust the stepper position based on Position Loop Error (0x35) . When this bit is set, the gain value Maximum Velocity Adjustment (0xE6) is multiplied by the Position Loop Error (0x35) , and the result is a velocity that is added to the microstepping position.
						2-15	Reserved.
0xF0	0xF1	0x4F0	0x2195:1	RF	U16	Debounce Time For Input 0. Units: ms.	
0xF1	0xF2	0x4F1	0x2195:2	RF	U16	Debounce Time For Input 1. Units: ms.	
0xF2	0xF3	0x4F2	0x2195:3	RF	U16	Debounce Time For Input 2. Units: ms.	
0xF3	0xF4	0x4F3	0x2195:4	RF	U16	Debounce Time For Input 3. Units: ms.	
0xF4	0xF5	0x4F4	0x2195:5	RF	U16	Debounce Time For Input 4. Units: ms.	
0xF5	0xF6	0x4F5	0x2195:6	RF	U16	Debounce Time For Input 5. Units: ms.	
0xF6	0xF7	0x4F6	0x2195:7	RF	U16	Debounce Time For Input 6. Units: ms.	
0xF7	0xF8	0x4F7	0x2195:8	RF	U16	Debounce Time For Input 7. Units: ms.	
0xF8	0xF9	0x4F8	0x2195:9	RF	U16	Debounce Time For Input 8. Units: ms.	
0xF9	0xFA	0x4F9	0x2195:10	RF	U16	Debounce Time For Input 9. Units: ms.	
0xFA	0xFB	0x4FA	0x2195:11	RF	U16	Debounce Time For Input 10. Units: ms.	
0xFB	0xFC	0x4FB	0x2195:12	RF	U16	Debounce Time For Input 11. Units: ms.	
0xFC	0xFD	0x4FC	0x2195:13	RF	U16	Debounce Time For Input 12. Units: ms.	
0xFD	0xFE	0x4FD	0x2195:14	RF	U16	Debounce Time For Input 13. Units: ms.	
0xFE	0xFF	0x4FE	0x2195:15	RF	U16	Debounce Time For Input 14. Units: ms.	
0xFF	0x100	0x4FF	0x2195:16	RF	U16	Debounce Time For Input 15. Units: ms.	
0x100	0x101	0x500	0x2184	RF	U32	CANopen Limit Status Mask. This parameter defines which bits in the Drive Event Status Register (0xA0) can set the limit bit (bit 11) of the CANopen Status Word (CANopen index 0x6041 as described in the <i>CANopen Programmer's Manual</i>). If a Drive Event Status Register (0xA0) bit and its corresponding Limit Mask bit are both set, then the CANopen Status Word limit bit is set. If all selected Drive Event Status	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description					
						Register (0xA0) bits are clear, then the limit bit is clear.					
0x101	0x102	0x501	0x2197	R*	INT16	Network Address Switch Value. This gives the current state of the CAN address switch. For drives without a switch, the value returned is undefined.					
0x102	0x103	0x502	0x21B4	R*	INT16	Network Status Word. Bit-mapped as follows:					
						CANopen					
						Bits	Meaning				
						0-1	CANopen node status. This field will take one of the following values:				
							Value	Status			
							0	The CANopen interface is disabled.			
							1	Stopped mode.			
							2	Preoperational mode.			
						3	Operational mode.				
						4	Set if the CANopen SYNC message is missing.				
						5	Set on a CANopen guard error.				
						8	Set if the CAN port is in 'bus off' state.				
						9	Set if the CAN port is in 'transmit error passive' state.				
						10	Set if the CAN port is in 'receive error passive' state.				
						11	Set if the CAN port is in 'transmit warning' state.				
						12	Set if the CAN port is in 'receive warning' state.				
						DeviceNet					
						Bits	Meaning				
						0	Set if duplicate MAC ID check failed.				
						1	Set if device is online.				
						2	Set if at least one communication object timed out.				
						3	Set if at least one communication object has been established.				
						4-7	Reserved.				
						8-14	Same bit mapping as for CANopen.				
						15	Always set for DeviceNet.				
						MACRO					
						0	Set if the MACRO network is detected.				

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						1	Set if the drive is being disabled by the MACRO master.
						2	Set if the MACRO network has been broken (i.e., once detected but now gone).
						3	Set on heartbeat error.
						4-15	Reserved.
0x103	0x104	0x503	0x21B1	F	U32	Input Pin Mapping For Node ID Selection. When Network Node ID Configuration (0xC1) indicates that 1 or more input pins will be used to select the node ID, this parameter is used to map input pins to ID bits.	
						Bits	Meaning
						0-3	Identify the general-purpose input pin associated with ID bit 0.
						4-7	Identify the general-purpose input pin associated with ID bit 1.
						8-11	Identify the general-purpose input pin associated with ID bit 2.
						12-15	Identify the general-purpose input pin associated with ID bit 3.
						16-19	Identify the general-purpose input pin associated with ID bit 4.
						20-23	Identify the general-purpose input pin associated with ID bit 5.
						24-27	Identify the general-purpose input pin associated with ID bit 6.
						28-30	Reserved.
						31	Set to enable this register. Clear to use default mapping.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description												
						If bit 31 is zero, then a default bit mapping is used and the rest of this register is ignored. The default bit mapping uses the top N input pins and maps them such that the high numbered pins are used for higher numbered bits in the ID. For example; the Accelnet panel drive has 12 general-purpose input pins (0 to 11). If 3 of these pins are used for ID configuration and the default mapping is used, then the highest 3 pins (9, 10 and 11) will be used for the ID. In this case, pin 9 will be bit 0, pin 10 will be bit 1, and pin 11 will be bit 2. If bit 31 is set, then the rest of this register will be used to define which input pin will be assigned to which bit of the ID. The input pins are numbered from 0 to 15 and each nibble of the register gives the input pin number associated with one bit of the ID. For example, if three input pins are configured for address selection and the mapping register is set to 0x80000012, then input pin 2 will be used for ID bit 0, input pin 1 will be used for ID bit 1, and input pin 0 will be used for ID bit 2. Note that the CAN node ID is calculated at startup only. The input pins assigned to the node ID will be sampled once during power up and used to calculate the ID. These pins may be assigned other uses after power up if necessary.												
0x104	0x105	0x504	0x21C4	RF	INT16	Algorithmic Phase Initialization Config. See code 5 of Commutation Mode (0xB2). This parameter is bit-mapped as follows: <table><tr><th>Bits</th><th>Meaning</th></tr><tr><td>0</td><td>If clear, use algorithmic phase initialization. If set, force the phase angle to zero degrees.</td></tr><tr><td>1</td><td>If set, increment the initial phase angle by 90 degrees on each failed attempt.</td></tr><tr><td>2</td><td>If set, use Motor Hall Offset (0x4F) as the initial angle for the first phase initialization attempt. If clear, the first phase angle is zero.</td></tr><tr><td>3</td><td>Ignore limit switches during phase initialization if the switch is configured as trajectory based. Available in Feature set C only.</td></tr><tr><td>4-15</td><td>Reserved.</td></tr></table>	Bits	Meaning	0	If clear, use algorithmic phase initialization. If set, force the phase angle to zero degrees.	1	If set, increment the initial phase angle by 90 degrees on each failed attempt.	2	If set, use Motor Hall Offset (0x4F) as the initial angle for the first phase initialization attempt. If clear, the first phase angle is zero.	3	Ignore limit switches during phase initialization if the switch is configured as trajectory based. Available in Feature set C only.	4-15	Reserved.
Bits	Meaning																	
0	If clear, use algorithmic phase initialization. If set, force the phase angle to zero degrees.																	
1	If set, increment the initial phase angle by 90 degrees on each failed attempt.																	
2	If set, use Motor Hall Offset (0x4F) as the initial angle for the first phase initialization attempt. If clear, the first phase angle is zero.																	
3	Ignore limit switches during phase initialization if the switch is configured as trajectory based. Available in Feature set C only.																	
4-15	Reserved.																	
0x105	0x106	0x505	0x2360	RF	U16	Camming Configuration. Bit-mapped as follows. For more information, see the <i>Camming User Guide</i>. <table><tr><th>Bits</th><th>Description</th></tr><tr><td>0-3</td><td>ID Number of the Cam Table to use (0-9).</td></tr><tr><td>4</td><td>Reserved.</td></tr></table>	Bits	Description	0-3	ID Number of the Cam Table to use (0-9).	4	Reserved.						
Bits	Description																	
0-3	ID Number of the Cam Table to use (0-9).																	
4	Reserved.																	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description		
						5	If set, exit table in forward direction.	
						6	If set, use the Camming Internal Generator. The internal generator runs at the constant velocity programmed in Camming Master Velocity (0x109) . If clear, use digital command input as configured in using 's CME 2 software camming controls or Input Pin States (0xA6) .	
						7	If set, run tables stored in RAM. If clear, use tables stored in the flash file system.	
						8-11	Input number to use as Cam Trigger. Note: a value of 0 selects IN1, 1 selects IN2, etc.	
						12-14	Cam Trigger type:	
							Value	Type
							0	None (Continuous): The active Cam Table is repeated continuously.
							1	Use Input, Edge: The active Cam Table begins executing on the rising edge of the input pin selected by bits 8-11.
							2	Use Input, Level: The active Cam Table will run as long as the input selected by bits 8-11 is high.
							3	Use Master (Secondary) Encoder Index: The active Cam Table is executed when the drive receives an index pulse from the Master encoder. Index pulses received during execution are ignored.
7	Never trigger. This can be used to stop a CAM currently in progress							
0x106	0x107	0x506	0x2361	RF	INT16	Camming delay, forward motion. This gives the delay (in master encoder counts) used when entering a cam table in the forward direction. Units: master command counts.		
0x107	0x108	0x507	0x2362	RF	INT16	Camming delay, reverse motion. This gives the delay (in master encoder counts) used when entering a cam table in the reverse direction. Units: master command counts.		
0x108	0x109	0x508	None	R	INT16	Writing any value to this parameter will cause any CANopen PDO objects configured with type code 254 to be sent. This parameter is primarily useful for triggering a PDO from within a CVM program. Reading this parameter does not return any useful information.		
0x109	0x10A	0x509	0x2363	RF	INT 32	Camming Master Velocity. Constant velocity of the Camming Internal Generator. Units: 0.1 counts/s.		
0x10A	0x10B	0x50A	0x2403	R*	INT 32	Captured Home Position. Provides the position that the axis was in when an input pin configured as a home switch input became active. Configured by setting bits in the Position Capture Control Register (0x6C) . Status of the captured data can be checked in the Position Capture Status Register (0x6D) . Reading this variable resets		

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						bits 4 & 7 of the Position Capture Status Register (0x6D) . Units: counts.
0x10B	0x10C	0x50B	0x2422	R*	U32	Firmware Version Number (extended). The upper 16 bits give the same major/minor version number as Firmware Version Number (0x94) . The lower 16 bits hold a release number (upper byte) and a reserved byte (lower).
0x10C	0x10D	0x50C	0x1017	RF	U16	CANopen Heartbeat Time. The frequency at which the drive will produce heartbeat messages. This parameter may be set to zero to disable heartbeat production. Note that only one of the two node-guarding methods may be used at once. If the Heartbeat Time is non-zero, then the heartbeat protocol is used regardless of the settings of the CANopen Node Guarding Time (0x10D) and CANopen Node Guarding Time (0x10D) . Units: ms.
0x10D	0x10E	0x50D	0x100C	RF	U16	CANopen Node Guarding Time. This parameter gives the time between node-guarding requests that are sent from the CANopen master to this drive. The drive will respond to each request with a node-guarding message indicating the internal state of the drive. If the drive has not received a node-guarding request within the time period defined by the product of the Node Guarding Time and the CANopen Node Guarding Life Time Factor (0x10E) , the drive will treat this lack of requests as a fault condition. Units: ms.
0x10E	0x10F	0x50E	0x100D	RF	U8	CANopen Node Guarding Life Time Factor. This object gives a multiple of the CANopen Node Guarding Time (0x10D) . The drive expects to receive a node-guarding request within the time period defined by the product of the Guarding Time and the Lifetime Factor. If the drive has not received a node-guarding request within this time period, it treats the lack of requests as a fault.
0x10F	0x110	0x50F	0x2325	R	INT 32	Registration Offset For Pulse & Direction Mode. When running in pulse & direction mode (Desired State (0x24) = 23), this parameter may be used to inject an offset into the master position. The offset will immediately be cleared once it has been applied to the master position, so this parameter will normally be read back as zero when running in pulse and direction mode 23.
0x110	0x111	0x510	0x2404	R	INT 32	Time Stamp of Last High Speed Position Capture. If high speed position capture is enabled, this parameter gives the time of the last capture. Setting this parameter causes the drive to calculate its position at the set time if

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						position capture is enabled and the time is recent enough for the data to be available. The calculated position may be read from <i>Captured Position for High Speed Position Capture</i> . Units: microseconds.	
0x111	0x112	0x511	0x2405	R*	INT 32	Captured Position for High Speed Position Capture. Units: counts.	
0x112	0x113	0x512	0x2242	R	INT 32	Position Encoder Position. This is also the passive load position when used in passive mode. Units: counts.	
0x113	0x114	0x513	0x1015	RF	INT16	CANopen emergency inhibit time. Units: milliseconds.	
0x114	0x115	0x514	0x2381:5	RF	U16	Velocity loop Drain (integral bleed). Modifies the effect of velocity loop integral gain. The higher the Vi Drain value, the faster the integral sum is lowered. Range: 0 to 32K Default: 0.	
0x115	0x116	0x515	0x2010	R	5 Words	Trajectory buffer access. This object can be used to load data into the drive's internal trajectory buffer, or send commands used to control the buffer. The trajectory buffer holds trajectory segments used in PVT mode. Data passed to the parameter consists of a 16-bit command code, followed by up to two 32-bit parameters. The first word passed to this parameter is bit-mapped. The data contained in this word identifies this access as either a buffer command, or a trajectory segment to be loaded into the buffer. If the most significant bit of the first word is set, then the write is treated as a command code. In this case no additional data is passed and the first word is formatted as follows:	
						Bits	Description
						0-7	Command data.
						8-9	Command code.
						10-14	Reserved.
						15	Always set for buffer commands.
						The following command codes are supported:	
						Code	Description
						0	Clear the buffer and abort any move in progress.
						1	Pop the N most recently sent segments off the buffer. PVT profiles will continue to run as long as the buffer doesn't underflow. The number of segments to pop (N) is passed in the command data area. If there are less than N segments on the buffer, this acts the same as a buffer clear, except that the profile is not stopped except by underflow.
						To write data to the trajectory buffer, the MSB of the first word should be clear. In this case the first word is formatted as follows:	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						Bits Description
						0-7 Segment time in milliseconds.
						8-11 Reserved.
						12 Set for relative positions; clear for absolute positions.
						13-14 Reserved.
						15 Always zero for data writes.
						When writing a new PVT segment to the trajectory buffer, this first word is always followed by a 32-bit position value. The position is specified in units of encoder counts and can be interpreted as either absolute or relative based on bit 12 of the command word.
						Optionally, the position can be followed by a 32-bit velocity value. Velocity is specified in units of 0.1 encoder counts / second. If the velocity is supplied, then the drive will use cubic polynomial interpolation between points when running the trajectory (PVT mode). If velocity is not supplied, then linear interpolation will be used (PT mode). It's legal to mix PVT and PT segments within the same move.
						Reading this parameter always returns three words of status information about the trajectory buffer.
						The first returned word is formatted as follows:
						Bits Description
						0-7 Number of free locations in the trajectory buffer.
						8-15 Reserved.
						The second two words are reserved for future use.
0x116	0x117	0x516	0x605A	RF	INT16	CANopen quick stop option code.
0x117	0x118	0x517	0x605B	RF	INT16	CANopen shutdown option code.
0x118	0x119	0x518	0x605C	RF	INT16	CANopen disable option code.
0x119	0x11A	0x519	0x605D	RF	INT16	CANopen halt option code.
0x11A	0x11B	0x51A	0x2080	F*	U32	Drive scaling configuration. Defines the units used for current and voltage readings from the drive:

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						Bits	Description
						0-1	Identify units for current readings:
							0 0.01 A
							1 0.001 A
							2 0.0001 A
							3 0.00001 A
						2-7	Reserved
						8-9	Identify units for voltage readings:
							0 0.1 V
							1 0.01 V
							2 0.001 V
							3 0.0001 V
						10-31	Reserved
0x120	0x121	0x520	0x2384:25	R*	INT16	Returns the number of axis implemented by this drive.	
0x121	0x122	0x521	0x21B3	RF	INT16	Network options. Configures the drive's network. The details of its meaning depend on the type of network implemented in the drive.	
						CANopen	
						Bits	Meaning
						0	Must be clear to select CANopen networking.
						1-15	Reserved.
						DeviceNet	
						Bits	Meaning
						0	Must be set to select DeviceNet networking.
						1-15	Reserved.
						MACRO	
						Bits	Meaning
						0	If set, position data sent over the MACRO network is shifted up 5 bits for compatibility with Delta-Tau controllers.
						1	If set, the drive will be disabled on startup until it is enabled through the MACRO interface. If clear, the drive can be used without the MACRO interface connected until it starts receiving MACRO messages.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						2 If set, return the primary encoder index state (high/low) in the home status bit of the MACRO status word. If clear, the state of any general purpose input configured as a home input will be used.
						3 If set, the drive will attempt to synchronize its current loop update period to the MACRO ring period. The ring period must be an integer multiple of the drive's PWM period (parameter 0x85).
						4-7 Identify the type of data sent from the drive in the first 16-bit MACRO register. 0=digital inputs, 1=secondary analog input 1, 2-15=reserved
						8-9 Identify the type of data sent in the second 16-bit MACRO register: 0 – primary analog input. 1 – Primary encoder reading. 2 – Secondary encoder reading. 3 – Pulse & direction hardware count.
						10-15 Reserved.
						EtherCAT
						Bits Meaning
						0 If set, disable some extra checks of the SYNC0 configuration which were added for improved network conformance.
						1 If set, the drive will follow the EtherCAT state machine even when running in a non EtherCAT mode of operation.
						2 If set, object 0x1002 is the bit-wise OR of all axes event status for multi-axis drives. If clear, 0x1002 is for axis 1 only.
						3-15 Reserved.
0x122	0x123	0x522	0x2384:26	F*	INT16	Internal maximum regen current. Units: mA.
0x123	0x124	0x523	0x2220	RF	INT32	Motor encoder wrap position. Actual motor position will wrap back to zero when this value is reached. Setting this value to zero disables this feature. Units: counts
0x124	0x125	0x524	0x2221	RF	INT32	Load encoder wrap position. Actual load position will wrap back to zero when this value is reached. Setting this value to zero disables this feature. Units: counts.
0x125	0x126	0x525	None	RF	INT16	Configures the MACRO drive's encoder capture circuit. This parameter is only used on MACRO drives. Bit-mapped as follows:
						Bits Meaning
						0-3 Type of capture to use.
						Value Description

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description		
							0	Capture on edge of encoder index.
							1	Capture using a general purpose input pin.
							2-15	Reserved.
						4-7	Input pin number to use if using capture type 1.	
						8	Active level; high if clear, low if set.	
						9	If set, capture is re-enabled immediately when the capture position is read (using I-variable 921). If clear, capture is only re-enabled on an explicit clear instruction.	
						10	If set, passive load encoder, if configured, will be captured. Passive load encoder currently only supports capture type 1 (general purpose input).	
						11-15	Reserved.	
0x126	0x127	0x526	0x2384:27	R*	INT16	FPGA firmware version number.		

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description		
0x127	0x128	0x527	0x2370	RF	U32	Gain scheduling configuration:		
						Bits	Meaning	
						0-2	Key parameter for gain scheduling.	
							Value	Description
							0	None. Setting the key parameter to zero disables gain scheduling.
							1	Use value written to Gain scheduling key parameter value (0x128) as the key.
							2	Use Instantaneous Commanded Velocity (0x3B) .
							3	Use Load Encoder Velocity (0x5F) .
							4	Use Commanded Position (0x2D) .
							5	Use Actual Position (0x17) .
							6-7	Reserved.
						3-7	Reserved.	
						8	If set, use the absolute value of key parameter for gain lookup.	
						9	If set, disable gain scheduling until the position encoder is referenced.	
10-31	Reserved.							
0x128	0x129	0x528	0x2371	R	INT32	Gain scheduling key parameter value. When gain scheduling is enabled, the current value of the key parameter is stored here. When this parameter is selected as the key parameter for gain scheduling, then it may be written to manually move through entries in the gain scheduling table.		
0x129	0x12A	0x529	0x2384:29	R	U32	Drive Hardware Options. Reserved for Controls use.		
0x12A	0x12B	0x52A	0x2222	F	U32	Motor encoder options. Used to specify various configuration options for the motor encoder. The mapping of option bits to function depends on the encoder type. Any bit not defined for a particular encoder should be considered reserved. Reserved bits should be set to zero to ensure compatibility with future firmware updates. Bit-mapped as follows:		
						Quadrature Encoder		
						Bits	Description	
						0	If set, ignore differential signal errors (if detected in hardware).	
						1	If set, select single ended encoder inputs (if available in hardware).	
						2	If set, ignore differential signal errors on index input only (if supported by hardware).	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						3	If set, don't use index input at all. Useful when the index input is being used by a different encoder interface.
						EnDat Encoder (Type 11)	
						Bits	Description
						0-5	Number of bits of single turn data available from encoder.
						8-12	Number of bits of multi-turn data available from encoder.
						16	Set if analog inputs are supplied by encoder.
						17	If set, use multi-mode port.
						18	If set, read position using EnDat 2.2 style commands rather than the default 2.1 style.
						20-23	Number of least significant bits of the encoder reading to discard.
						SSI Encoder (Type 12)	
						Bits	Description
						0-5	Number of bits of position data available. A value of zero is invalid for these bits.
						8-11	Number of extra bits sent with position data.
						12	If set, ignore the first bit of data sent by the encoder.
						13	If set, encoder outputs position data using Gray code.
						14	If set, pull clock low briefly after data (custom for Codechamp encoder).
						15	If set, data is sent LSB first.
						16-21	Encoder bit rate in 100 kHz units. If zero, default to 1 MHz.
						22	If set, use setting of encoder counts/rev to determine how many data bits to use.
						23	If set, extra status bits are before position data. If clear, extra status bits are after position data. Default is clear.
						24	If set, first bit sent is 'data valid' bit.
						BiSS (Type 13)	
						Bits	Description
						0-5	Number of bits of single turn data.
						8-12	Number of bits of multi-turn data.
						15	If set, assume the encoder position data wraps after the number of encoder counts programmed in parameter Motor Encoder Counts (0x62) .
						16	Set for mode-C encoder format.
						20	Set if encoder error and warning bits are active low.

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						21 Set if encoder status bits are sent before position data. Clear if status bits are sent after position data.
						22 Set if encoder error bit is transmitted before warning bit. Clear for warning bit sent first.
						24-26 Number of alignment bits (reserved bits sent before position info).
						28 Use multi-mode port if set. If clear use primary encoder interface.
						30 Set for 2.5 MHz baud rate, clear for 4 MHz baud rate.
						Tamagawa, Panasonic, Harmonic Drives, etc. (Type 14)
						Bits Description
						0-5 Number of bits of single turn data.
						8-12 Number of bits of multi-turn data.
						16-19 Number of LSB to discard from reading.
						20-22 Number of consecutive CRC errors to ignore before generating an error.
						24-27 Encoder sub-type (0=Tamagawa, 1=Panasonic absolute, 2=HD systems, 3=Panasonic Incremental, 4=Sanyo Denki, 5=Tamagawa Single Turn).
						28 Bit rate (set for 4 Mbit, clear for 2.5 Mbit).
						30 If set, treat encoder battery errors as warnings.
0x12B	0x12C	0x52B	0x2223	F	U32	Load Encoder Options. Same as 0x12A, but affects load or position encoder.
0x12C	0x129	0x52C	0x2384:28	R*	U32	Secondary firmware version for drives equipped with two processors.
0x12D	0x12E	0x52D	0x2109	RF	9 or 14 words	Analog Input Filter Coefficients. A bi-quad filter which acts on the analog reference input. 9 word parameter, see <i>Analog Input Filters</i> in the <i>CME 2 User Guide</i> . 14 word parameter (Plus product only), see Filter Coefficients (p. 73) .
0x12E	0x12F	0x52E	0x2224	R*	U32	Motor encoder status. This parameter gives additional status information for the encoder. Bits set in the status word are latched and cleared when the status value is read. The format of this status word is dependent on the encoder type. Many error bits are taken directly from encoder data stream. For a full description of what these error bits mean, please consult the encoder manufacturer. Quadrature Bits Description 0 Reserved. 1 Set on bad differential signal levels on any of the encoder inputs. EnDAT (Type 11) Bits Description

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						0	CRC error on data received from encoder.
						1	Failed to detect encoder connected to drive.
						2	Error bit on encoder stream is active.
						3	Encoder failed to respond to request for position.
						SSI (Type 12)	
						Bits	Description
						0-6	Fault flags returned from encoder.
						15	Encoder data invalid bit set.
						BiSS (Type 13)	
						Bits	Description
						0	CRC error on data received from encoder.
						1	Encoder failed to transmit data to amp.
						2	Error bit on encoder stream is active.
						3	Warning bit on encoder stream is active.
						4	Encoder transmission delay is too long.
						Tamagawa & Panasonic (Type 14)	
						Bits	Description
						0	Over speed error reported by encoder.
						1	Absolute position error reported by encoder.
						2	Counting error reported by encoder.
						3	Counter overflow reported by encoder.
						5	Multi-turn error reported by encoder.
						6	Battery error reported by encoder.
						7	Battery warning reported by encoder.
						8	Error bit 0 reported by encoder.
						9	Error bit 1 reported by encoder.
						10	Comm error 0.
						11	Comm error 1.
						15	CRC error on data received from encoder.
						Sanyo Denki & Harmonic Drives (Type 14)	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						Bits	Description
						0	Battery warning reported by encoder.
						1	Battery error reported by encoder.
						3	Over speed reported by encoder.
						4	Memory error reported by encoder.
						5	STERR reported by encoder.
						6	PSERR reported by encoder.
						7	Busy error reported by encoder.
						8	Memory busy reported by encoder.
						9	Over temperature.
						15	CRC error on data received from encoder.
0x12F	0x130	0x52F	0x2225	R*	U32	Load encoder status. Same as parameter 0x12E, but for the load encoder.	
0x130	0x131	0x530	0x2114	RF	INT16	RMS current calculation period. This sets the period over which the RMS current is calculated. If this value is set to zero, then the RMS current will be updated each time it is read for the period since the last read. In this case, the RMS current must be read at least once every 65536 current loop periods (about every 4 seconds) for the returned RMS values to be accurate. Units: milliseconds.	
0x131	0x132	0x531	0x2115	R*	INT16	RMS current over the period set in parameter 0x130. Units: 0.01 A.	
0x132	0x133	0x532	0x2116	R*	INT16	Running sum of user current limit in 0.01% units. (i.e., 0 to 10000).	
0x133	0x134	0x533	0x2117	R*	INT16	Running sum of amp current limit in 0.01% units. (i.e., 0 to 10000).	
0x134	0x135	0x534	0x21E0	RF	U32	Analog output configuration. This parameter sets the mode for the Analog output on drives so equipped.	
						Bits	Description
						0-3	Define the mode of the Analog output.
						16-17	Identify the axis associated with the Analog output.
						Mode	Description
						0	Manual configuration (set using parameter 0x135).
						1	Actual current of configured axis.
						2	Actual velocity of configured axis. The output is a +/- 5V range and is a ratio of the actual velocity to the velocity loop maximum velocity.
0x135	0x136	0x535	0x21E1	R	INT16	Analog output value. For drives that support an Analog output, this parameter sets	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
						the output value in mV units when the D/A is in manual mode. In other modes, the current value being output on the D/A can be read here.
0x136	0x137	0x536	0x2208	R*	INT16	Secondary analog reference value. Units: millivolts
0x137	0x138	0x537	0x2314	RF	INT16	Secondary analog reference offset. Offset for secondary analog reference input. Units: millivolts
0x138	0x139	0x538	0x2315	RF	INT16	Calibration offset for second analog reference input. It is factory-calibrated to give a zero reading for zero input voltage. Units: millivolts
0x139	0x140	0x539	0x219D	R	INT32	Status of drive safety circuit. This parameter allows the status of the safety circuit built into some drives to be queried. For drives without a safety circuit, this parameter is reserved.
						Bits Description
						0 Set when safety input 0 is preventing the drive from enabling.
						1 Set when safety input 1 is preventing the drive from enabling.
						8 This read/write bit can be used to force the 'drive is unsafe' output of the safety circuit to go active for testing purposes. Write 1 to force.
0x13A	0x13B	0x53A	0x2209	R*	INT16	Present voltage at analog motor temperature sensor. Units: millivolts.
0x13B	0x13C	0x53B	0x220A	RF	INT16	Limit for analog motor temperature sensor. If this parameter is set to zero, then the analog motor temperature sensor is disabled. If this parameter is set to a positive value, then a motor temperature error will occur any time the voltage on the motor temperature input exceeds this value (in millivolts). If this parameter is negative, then a motor temperature error will occur any time the voltage on the motor temperature input is lower than the absolute value of this limit in millivolts.
0x13C	0x13D	0x53C	0x2323	RF	INT16	Minimum PWM pulse width in microseconds. Used when running in PWM position mode. In this mode the PWM input pulse width is captured by the drive and used to calculate an absolute position using the following formula: $\text{pos} = ((\text{PW-MIN}) / (\text{MAX-MIN})) * \text{SCALE} + \text{OFFSET}$ where this parameter is the minimum pulse width (MIN), parameter 0x13D is the maximum pulse width (MAX), parameter 0xA9 is the scaling factor (SCALE) and parameter 0x10F is the offset (OFFSET).
0x13D	0x13E	0x53D	0x2324	RF	INT16	Maximum PWM pulse width used when running in PWM position mode. Units:

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
						microseconds.	
0x150	0x151	0x550	0x210A	RF	14 words	Second chained biquad filter on output of velocity loop, see Filter Coefficients (p. 73) .	
0x151	0x152	0x551	0x210B	RF	14 words	Third chained biquad filter on output of velocity loop. 14 word parameter, see Filter Coefficients (p. 73) .	
0x152	0x153	0x552	0x210C	RF	14 words	First chained biquad filter on input of current loop. 14 word parameter, see Filter Coefficients (p. 73) .	
0x153	0x154	0x553	0x210D	RF	14 words	Second chained biquad filter on input of current loop. 14 word parameter, see Filter Coefficients (p. 73) .	
0x154	0x155	0x554	0x2301	RF	INT32	Servo loop configuration. This parameter allows various parts of the drive servo loops to be enabled/disabled. Bit-mapped as follows:	
						Bits	Description
						0	If set, this disables the velocity loop gains. The velocity loop command feed forward gain (parameter 0x157) is still active as are the velocity loop output filters.
						1	If set, this enables the position loop I and D gains (parameters 0x155 and 0x156). If clear, these parameters are treated as zeros.
						2-31	Reserved for future use.
0x155	0x156	0x555	0x2382:5	RF	INT16	Position loop integral gain (KI).	
0x156	0x157	0x556	0x2382:6	RF	INT16	Derivative gain for position loop (KD).	
0x157	0x158	0x557	0x2381:6	RF	INT16	Velocity loop command feed forward. The input command (after limiting) to the velocity loop is scaled by this value and added in to the output of the velocity loop.	
0x158	0x159	0x558	0x2382:7	RF	INT16	Integral drain for position loop.	
0x159	0x15A	0x559	0x6007	RF	INT16	Abort option code for CANopen / EtherCAT drives.	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
0x15A	0x15B	0x55A	0x2198	RF	U32	I/O options. This parameter is used to configure optional features of the general purpose I/O.
						Bits Description
						0-3 For AEM/APM, these bits determine whether several I/O pins are used as a serial interface for expanded I/O features, and if so how they are configured. 0 – normal I/O 1 – AEM/APM development board LEDs and address switches. 2 – LEDs wired the same as the developer's kit board, but using separate red & green LEDs for the network status.
						4-31 Reserved for future use.
0x15B	0x15C	0x55B	0x2199	F	INT16	Motor brake enable delay time (milliseconds). This parameter gives a delay between enabling the drive PWM outputs and releasing the brake. Positive values mean the PWM is enabled first and then the brake is released N milliseconds later. Negative values cause the brake to be released before PWM outputs are enabled.
0x15C	0x15D	0x55C	0x219A	R*	U32	32-bit version of parameter 0xA6.
0x15D	0x15E	0x55D	0x219B	R*	U32	32-bit version of parameter 0xAA.
0x15E	0x15F	0x55E	0x219C	RF	U32	32-bit version of parameter 0xA5.
0x160- 0x167	0x161- 0x168	0x560- 0x567	0x2192:17	RF	U16	Input pin config for general purpose input 16. See Input 0 Configuration (0x78)
			0x2192:18	RF	U16	Input pin config for general purpose input 17. See Input 0 Configuration (0x78)
			0x2192:19	RF	U16	Input pin config for general purpose input 18. See Input 0 Configuration (0x78)
			0x2192:20	RF	U16	Input pin config for general purpose input 19. See Input 0 Configuration (0x78)
			0x2192:21	RF	U16	Input pin config for general purpose input 20. See Input 0 Configuration (0x78)
			0x2192:22	RF	U16	Input pin config for general purpose input 21. See Input 0 Configuration (0x78)
			0x2192:23	RF	U16	Input pin config for general purpose input 22. See Input 0 Configuration (0x78)
			0x2192:24	RF	U16	Input pin config for general purpose input 23. See Input 0 Configuration (0x78)
0x170- 0x177	0x171- 0x178	0x570- 0x577	0x2195:17	RF	U16	Debounce time for general purpose input 16.
			0x2195:18	RF	U16	Debounce time for general purpose input 17.
			0x2195:19	RF	U16	Debounce time for general purpose input 18.

Parameter Dictionary	Parameters																																					
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			0x2195:23	RF	U16	Debounce time for general purpose input 22.																																
			0x2195:24	RF	U16	Debounce time for general purpose input 23.																																
0x180	0x181	0x580	0x2326	RF	U32	UV configuration. This parameter is used to configure the drive when running in UV mode (desired state 5). Bit-mapped as follows (undocumented bits are reserved for future use): <table><tr><td>Bits</td><td>Meaning</td></tr><tr><td>0-1</td><td>Define the source of the UV command inputs:<table><tr><td>Value</td><td>Description</td></tr><tr><td>0</td><td>PWM inputs.</td></tr><tr><td>1</td><td>Analog reference inputs (for drives with two analog reference pins).</td></tr><tr><td>2</td><td>Analog encoder inputs.</td></tr><tr><td>3</td><td>Directly set over the serial/network interface.</td></tr></table></td></tr><tr><td>2-7</td><td>Reserved.</td></tr><tr><td>8-9</td><td>Define the format of the UV inputs:<table><tr><td>Value</td><td>Description</td></tr><tr><td>0</td><td>120 degree current commands.</td></tr><tr><td>1</td><td>90 degree current commands.</td></tr><tr><td>2</td><td>Angle/Magnitude form. U input gives magnitude, V gives angle.</td></tr></table></td></tr><tr><td>10-15</td><td>Reserved.</td></tr><tr><td>16</td><td>If set, the value of the Motor Hall Offset (0x4F) is added to the UV angle</td></tr><tr><td>17</td><td>If set, the drive will use field oriented control. Normally FOC is disabled in UV mode due to the ambiguity of the phase angle with zero inputs. This is best used when running in angle/magnitude format.</td></tr></table>	Bits	Meaning	0-1	Define the source of the UV command inputs: <table><tr><td>Value</td><td>Description</td></tr><tr><td>0</td><td>PWM inputs.</td></tr><tr><td>1</td><td>Analog reference inputs (for drives with two analog reference pins).</td></tr><tr><td>2</td><td>Analog encoder inputs.</td></tr><tr><td>3</td><td>Directly set over the serial/network interface.</td></tr></table>	Value	Description	0	PWM inputs.	1	Analog reference inputs (for drives with two analog reference pins).	2	Analog encoder inputs.	3	Directly set over the serial/network interface.	2-7	Reserved.	8-9	Define the format of the UV inputs: <table><tr><td>Value</td><td>Description</td></tr><tr><td>0</td><td>120 degree current commands.</td></tr><tr><td>1</td><td>90 degree current commands.</td></tr><tr><td>2</td><td>Angle/Magnitude form. U input gives magnitude, V gives angle.</td></tr></table>	Value	Description	0	120 degree current commands.	1	90 degree current commands.	2	Angle/Magnitude form. U input gives magnitude, V gives angle.	10-15	Reserved.	16	If set, the value of the Motor Hall Offset (0x4F) is added to the UV angle	17	If set, the drive will use field oriented control. Normally FOC is disabled in UV mode due to the ambiguity of the phase angle with zero inputs. This is best used when running in angle/magnitude format.
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0x181	0x182	0x581	0x2327	R	INT16	U input when running in UV mode. This parameter can be used to read the calculated U value, or to set a U value when the UV inputs are being directly set over the serial/network interface.																																
0x182	0x183	0x582	0x2328	R	INT16	V input when running in UV mode.																																
0x183	0x184	0x583	0x2329	R	INT16	Raw counter value from pulse & direction input hardware. This can be read when running in any mode, not just pulse & direction modes. This parameter can be written also, but should not be written when the amp is being driven by the pulse & direction inputs. Writing in that mode will cause the amp to treat the change in the counter as real pulse inputs resulting in possible unexpected motion.																																

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description														
0x184	0x185	0x584	0x2254	RF	8 to 40 Words	Input shaping filter. This filter is used to modify the trajectory before its input to the position loop. This can be used to compensate for low frequency resonances in the load. The parameter is an array of 32-bit values. The first four values are used to store information about the input shaping filter (filter type, frequency, etc.) and are mostly unused by the firmware. The only exception is that the MSB of the first word should not be set to ensure compatibility with future firmware versions. The remaining 32-bit values are pairs of IEEE floating point values. Each pair defines a time (first value) and an impulse amplitude (second value). Up to eight pairs may be passed for up to 8 impulses in the input shaping filter. The time values are specified in seconds and must be >= 0.0. The impulse values are unit-less and must have an absolute magnitude of < 16.0.														
0x185	0x186	0x585	0x2160	R	U32	Output compare configuration. For a detailed description of the output compare function, please see the Setting Outputs at Position application note describing it.														
0x186	0x187	0x586	0x2161	R	U32	Output compare status.														
0x187	0x188	0x587	0x2162	R	INT32	Output compare value 1.														
0x188	0x189	0x588	0x2163	R	INT32	Output compare value 2.														
0x189	0x18A	0x589	0x2164	R	INT32	Output compare increment.														
0x18A	0x18B	0x58A	0x2165	R	INT32	Output compare pulse width.														
0x18B	0x18C	0x58B	0x2255	RF	INT32	Trajectory options. This parameter is used to modify the behavior of some trajectory modes. Its interpretation depends on the trajectory mode being used. The following trajectory modes currently make use of this parameter: <table><tr><th colspan="2">EtherCAT CSP mode:</th></tr><tr><th>Bits</th><th>Description</th></tr><tr><td>0-7</td><td>Number of extra loop cycles to extrapolate trajectories if input data from the master is not received.</td></tr><tr><td>8-15</td><td>Reserved.</td></tr><tr><td>16</td><td>If set, jump to quick stop mode if master data is not received within the number of cycles set in bits 0-7.</td></tr><tr><td>17</td><td>If set, and the interpolation time object (0x60C2) is non-zero, then the calculated velocity will be filtered, and a trajectory acceleration will also be calculated. If not set, velocity is unfiltered and acceleration is not calculated (zero).</td></tr><tr><td>18-31</td><td>Reserved.</td></tr></table>	EtherCAT CSP mode:		Bits	Description	0-7	Number of extra loop cycles to extrapolate trajectories if input data from the master is not received.	8-15	Reserved.	16	If set, jump to quick stop mode if master data is not received within the number of cycles set in bits 0-7.	17	If set, and the interpolation time object (0x60C2) is non-zero, then the calculated velocity will be filtered, and a trajectory acceleration will also be calculated. If not set, velocity is unfiltered and acceleration is not calculated (zero).	18-31	Reserved.
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ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description	
0x18C	0x18D	0x58C	0x21A1	RF	U32	I/O extension configuration for modules. This parameter is used to configure the I/O extension feature on plus modules which support it. For a detailed description of this I/O extension feature, see the Extending Plus Module I/O application note.	
						Bits	Description
						0-4	Number of bits to transfer less 1 (e.g., set to 19 to transfer 20 bits).
						5-7	Reserved.
						8	Set to enable SPI I/O extension feature. Clear for LED/Switch interface.
						9	Automatically restart transmission if set.
						10	Leave the CS line low after transfer if set.
						11	Reserved.
						12	Clock polarity setting.
						13	Data phase setting.
						14-15	Reserved.
						16-23	Clock period (100ns units).
						24-31	Reserved.
0x18D	0x18E	0x58D	0x21A2	R	INT32*	I/O extension transmit data. Data to be transferred over the SPI port is sent immediately after being written to this parameter. *Data size is variable dependent on drive configuration. Refer to the Extending Plus Module I/O application note.	
0x18E	0x18F	0x58E	0x21A3	R	INT32*	I/O extension receive data. Data received from the SPI port can be read from this parameter. *Data size is variable dependent on drive configuration. Refer to the Extending Plus Module I/O application note.	
0x18F	0x190	0x58F	0x220B	RF	INT16	Encoder sine offset. This is set in A/D units and only used with resolvers and servo-tube motors. It gives an offset which is added to the encoder sine signal before calculating position.	
0x190	0x191	0x590	0x220C	RF	INT16	Encoder cosine offset. Like 0x18F, but for encoder cosine signal.	
0x191	0x192	0x591	0x220D	RF	U16	Encoder cosine scaling factor. Used by the resolver & servo tube encoder calculations. This scaling factor is used to adjust the cosine signal amplitude so that it's the same as the sine signal amplitude. If set to zero, both the scaling and offsets (0x18F,0x190) will be ignored. If non-zero the cosine is scaled by N/32768 where N is the value of this parameter.	

ASCII	DvcNet	MACRO	CAN/ECAT IDX:SUB	Bank	Type	Description
0x192	0x193	0x592	0x2226	RF	U32	Motor encoder calibration settings. The meaning of this parameter is dependent on the encoder type.
0x193	0x194	0x593	0x2227	RF	U32	Load encoder calibration settings. Same as 0x192, but applied to the load encoder.
0x194	0x195	0x594	0x232A	R*	INT16	PWM input duty cycle. This parameter can be used to read the duty cycle of the PWM input. The returned 16-bit value gives the duty cycle in the range +/-32k. Parameter 0xA8 is used to configure the PWM input.
0x195	0x196	0x595	0x2123	RF	INT32	Jerk value to use during trajectory aborts. If this is zero, then the abort will be calculated without any jerk limits. Units are 100 counts / s ³ .
0x196	0x197	0x596	0x220E	R*	INT32	Returns the magnitude squared of the analog encoder signals (sin*sin + cos*cos)
0x197	0x198	0x597	0x2378	RF	INT16	Cross coupling KP gain. On dual axis drives this gain is applied to the difference in position error of the two axes.
0x198	0x199	0x598	0x2379	RF	INT16	Cross coupling KI gain.
0x199	0x19A	0x599	0x237A	RF	INT16	Cross coupling KD gain.
0x19A	0x19B	0x59A	0x220F	RF	5 words	Steinhart constants for motor analog motor temperature sensor.
0x19B	0x19C	0x59B	0x2384:30	F*	INT16	Current at which minimum PWM deadtime is used.
0x19C	0x19D	0x59C	0x2406	R*	INT32	High speed capture for passive load encoder.
0x19D	0x19E	0x59D	0x2142	RF	INT16	Open motor wiring check current. If parameter 0x15B is greater than zero, then during that time period on enable this current will be applied to the motor wiring to check that the motor is connected. If the programmed current cannot be applied to the motor, then a motor disconnected fault will be flagged.
0x19E	0x19F	0x59E	0x6066	RF	U16	Position error timeout. The time that a position error must persist before it is triggered. Units: ms
0x1A0	0x1A1	0x5A0	0x2193:9	RF	See text	Output 9 Configuration. See Output 0 Configuraton (0x70) .
0x1A1	0x1A2	0x5A1	0x2193:10	RF	See text	Output 10 Configuration. See Output 0 Configuraton (0x70) .
0x1A2	0x1A3	0x5A2	0x2193:11	RF	See text	Output 11 Configuration. See Output 0 Configuraton (0x70) .
0x1A3	0x1A4	0x5A3	0x2193:12	RF	See text	Output 12 Configuration. See Output 0 Configuraton (0x70) .

CHAPTER

3: FILTER COEFFICIENTS

There are several drive parameters which are used to define filters. These filters are implemented as generic biquadratic filter structures. Filters of this type implement the following formula to transform the input parameter $x(n)$ at time n to an output parameter $y(n)$:

$$y(n) = b0 \cdot x(n) + b1 \cdot x(n-1) + b2 \cdot x(n-2) + a1 \cdot y(n-1) + a2 \cdot y(n-2)$$

The values $a1$, $a2$, $b0$, $b1$, $b2$ are constants known as filter coefficients. They define the type of filter being implemented.

The values passed to these drive filter parameters are used to define the filter coefficients. The formatting of these parameters varies depending on the drive family being interfaced to.

All first generation drives use 16-bit integer math to implement their filters internally. Filter coefficients are given as 16-bit signed integer values. To increase the resolution of these coefficients, an additional unsigned scaling coefficient (k) is also specified. The actual filter formula used within these drives is as follows:

$$y(n) = K \cdot (b0 \cdot x(n) + b1 \cdot x(n-1) + b2 \cdot x(n-2) + a1 \cdot y(n-1) + a2 \cdot y(n-2)) / 32768 / 4096$$

To set the filter coefficients on drives of this category, 9 words of parameter data are passed. The first three words of data are informational parameters which are used by the CME software to describe the filter. If the upper 3 bits of the first word are all set then the filter will be disabled. Other than that, the first three words of data are not used in any way by the firmware. These three words are reserved for CME use.

Word	Description
1	Filter info. Set to 0xFFFF to disable the filter. Otherwise, reserved for CME use.
2	Filter info. Reserved for CME use.
3	Filter info. Reserved for CME use.
4	b2 coefficient.
5	b1 coefficient.
6	b0 coefficient.
7	a2 coefficient.
8	a1 coefficient.
9	K scaler.

For the plus family of drives (Xenus+, AEM, etc), a new format is used to describe the biquad filter coefficients. These drives include the ability to design the filters in firmware using the Cephes filter design library.

Filters on this family of drives are calculated internally using 32-bit IEEE floating point coefficients. The format of the parameter information passed when setting filter parameters on these drives consists of an array of up to fourteen 16-bit words. The first 4 words describe the filter,

and the remaining 10 words give the filter coefficients as 32-bit IEEE floating point values. The filter coefficient words are optional and are only necessary if the firmware is not calculating the coefficients internally.

Word	Description	
1	Bits	Usage
	0-3	Filter family.
	4	If set, filter will not be designed. Always set by firmware after successfully designing the filter. This prevents the filter from being redesigned when copied from flash at startup.
	5-7	Reserved.
	8	Number of poles – 1 (i.e. 0 for single pole, 1 for two pole).
	9-12	Reserved.
	13-15	Filter type.
	All reserved bits should be set to zero. The filter family should be one of the following values:	
	0	Custom biquad filter. Coefficients must be passed; the firmware will not design the filter.
	1	Butterworth filter.
	2	Chebyshev filter.
	3	Elliptic filter.
	4-15	Reserved.
	The filter type should be one of the following:	
	0	Custom biquad filter. Coefficients must be passed; the firmware will not design the filter.
	1	Low pass.
	2	High pass.
	3	Band reject (notch).
	4	Band pass.
	5-6	Reserved
	7	Disabled. The filter will have no effect in the system.
	If legal values are passed for the filter type and family, the firmware will attempt to design the specified filter and fill in the coefficient values itself. The firmware is capable of calculating 1 or 2 pole low pass or high pass filters. For notch and band	

Word	Description	
	pass filters the firmware can only calculate a two pole filter. For these filter types, bit 8 must be set.	
2	This word gives the cut off frequency (in Hz) for low pass and high pass filters. For notch and band pass filters this gives the first filter frequency.	
3	This word gives the second filter frequency (in Hz) for notch and band pass filters.	
4	Bits	Usage
	0-7	Rp given in 0.1 dB units.
	8-15	Rs given in dB units.
	Rp is the pass band ripple. This parameter is only used for Chebychev and Elliptic filters. Rs is only used with elliptic filters. It defines the stop band as Rs dB down from the peak value in the pass band.	
5-6	Coefficient a1. All filter coefficients are passed as 32-bit IEEE floating point numbers. The upper 32-bits should be passed first. If the firmware designs the filter then the coefficients will be filled in by the firmware and need not be passed.	
7-8	Coefficient a2.	
9-10	Coefficient b0.	
11-12	Coefficient b1.	
13-14	Coefficient b2.	